

MDS211 Nervous System — Past Paper Recreation

Lectures 1–16 (Exam 1 / FA-SA1) · 251 Questions · Single Best Answer

Compiled from the MDS211 past-paper archive spanning multiple cohorts (MED33–VCKMD36), covering Lectures 1–16 (Formative/Summative Exam 1). All 251 recorded questions are included, de-duplicated where the source explicitly flagged repeats. The answer key has been fully re-verified against source neuroscience material rather than the original automated colour-detection extraction — 36 answers that were factually wrong have been corrected, and 27 remain flagged where the question is genuinely ambiguous, references an image/table not captured in the text, or has more than one defensible answer.

Lecture 1 — Organization of the CNS (3 questions)

1. Concerning the parts of Central Nervous System, which of the following items IS NOT included?
 - A. Pineal gland
 - B. Basal ganglia
 - C. Fourth ventricle
 - D. Trigeminal ganglion
 - E. Posterior pituitary gland
2. Concerning the parts of Central Nervous System, which of the following items IS NOT included?
 - A. Pineal gland
 - B. Basal ganglia
 - C. Fourth ventricle
 - D. Trigeminal ganglion
 - E. Posterior pituitary gland
3. When we wear a ring or watch, our body receives less stimulation from the central nervous system because of which receptor has adaptation?
 - A. Meissner's receptor
 - B. Pacinian receptor
 - C. Free nerve ending
 - D. Krause receptor
 - E. Peritrichial receptor (hair root plexus)

Lecture 2 — Neurohistology (Neuron & Neuroglia) (19 questions)

4. Which of the following compounds is transferred from astrocyte to neuron and use for ATP production during starvation?
 - A. Fumarate
 - B. Citrate
 - C. Pyruvate 5
 - D. Succinyl-CoA
 - E. Lactate

5. Myelin sheath wrapping nerve axons in the basal ganglia is produced by which of the following cells?
- A. Astrocyte
 - B. Schwann
 - C. Satellite
 - D. Oligodendrocyte
 - E. Microglia
6. Which of the following is the Glutamate transporter into astrocyte?
- A. MCT
 - B. DCT
 - C. IAATS
 - D. TCT
 - E. EAATs
7. If the structure that carry fast pain is damaged, which of the following can be described as the structure that carry fast pain?
- A. A- β fiber
 - B. Diameter about 13-20 micrometer fiber
 - C. Paleospinothalamic tract
 - D. Neo-lateral spinothalamic tract
 - E. Thickest myelin sheath
8. Which of the following compounds is transferred from astrocyte to neuron and use for ATP production during starvation?
- A. Citrate
 - B. lactate
 - C. fumarate
 - D. succinyl-CoA
 - E. pyruvate
9. Which of the following items IS NOT a component of the blood-brain-barrier?
- A. Oligodendrocyte
 - B. Astrocyte end-feet
 - C. Pericyte of capillary
 - D. Endothelial cell of capillary
 - E. Tight junction protein
10. If the structure the carry fast pain damage, which of the following can be described the structure that carry fast pain?
- A. A β fiber
 - B. Thickest myelin sheath
 - C. Paleospinothalamic tract
 - D. Neo-lateral spinothalamic tract
 - E. C fiber

- 11.** Which of the following cells is associated with the immune system in the CNS?
- A. Microglia cell
 - B. Schwann cell
 - C. Oligodendrocyte
 - D. Astrocyte
 - E. Ependymal cell
- 12.** Which part of the axon has dense voltage-gated channels needed for Action potential?
- A. Node of Ranvier
 - B. Myelin sheath
 - C. Soma
 - D. Axon terminal
 - E. Dendrite
- 13.** Which disease that immune system attacks and destroys myelin sheath?
- A. Alzheimer disease
 - B. Multiple sclerosis
 - C. Parkinson's disease
 - D. Myasthenia Gravis
 - E. Guillain-Barré syndrome
- 14.** What is correct about sensory ganglia and autonomic ganglia?
- A. Sensory ganglia are pseudounipolar, ANS ganglia are multipolar
 - B. sensory in PNS sympathetic in CNS
 - C. sensory ganglion schwann cell sympathetic oligodendrocytes
 - D. Both are found only in the CNS
 - E. Both contain multipolar neurons exclusively
- 15.** Which is incorrect about the nervous system?
- A. Neuron has glycogen synthase
 - B. Astrocyte uptake glucose via GLUT1
 - C. Lactate is transferred by Astrocyte during brain starvation
 - D. Free radical is produced by oxidative phosphorylation in ETC in Neuron
 - E. Microglia are the resident CNS macrophage-like immune cells
- 16.** Which of the following is the glutamate transporter to astrocytes?
- A. MCT
 - B. DCT
 - C. IAATS
 - D. TCT
 - E. EAATS
- 17.** Which of the following is the Lactate transporter to astrocytes?
- A. DCT
 - B. MCT
 - C. IAATS
 - D. TCT
 - E. EAATS

18. The structure around the nerve fiber (1st layer: from innermost) is

- A. Epinurium
- B. Myelin sheath
- C. Neurolemma
- D. Endonurium
- E. Bundle

19. Which of these is not part of the BBB (Blood-Brain Barrier)?

- A. Pericyte
- B. Endothelium
- C. Astrocyte
- D. Ependymal cell
- E. Basement membrane

20. What is the 1st layer that covers an axon (from inside) in the PNS?

- A. Neurilemma
- B. Myelin sheath
- C. Endoneurium
- D. Perineurium
- E. Epineurium

21. Which statement is correct about a neuron?

- A. dendrite send electric signal away from soma
- B. axon specially in receptive function
- C. golgi apparatus term nissl body act ribosome...
- D. neuron has abundant of rER
- E. All neurons are unipolar in structure

22. The innermost structure around a single nerve fiber?

- A. Epinurium
- B. Myelin sheath
- C. Neurolemma
- D. Endonurium
- E. Bundle

Lecture 3 — Neuron Properties: RMP & Action Potential (10 questions)

23. A 40-year-old male presented with hand injury needs immediate suture. Lidocaine is given as a local anesthesia. What is the mechanism of lidocaine? [NL 2021]

- A. Activation of ligand-gated potassium channel
- B. Inhibition of nicotinic acetylcholine receptor
- C. Inhibition of voltage-gated sodium channel
- D. Inhibition of acetylcholine release
- E. Activation of muscarinic acetylcholine receptor

24. Hyperpolarization of the postsynaptic membrane in response to glycine or GABA is produced by the opening of. . .
- A. K^+ channels
 - B. Cl^- channels
 - C. Na^+ channels
 - D. Ca^{2+} channels
 - E. Mg^{2+} channels
25. A 35-year old male was given a lidocaine injection prior to incision and drainage of the abscesses () at right buttocks. Which of the following is the mechanism of action () of lidocaine?
- A. Block voltage-gated K^+ channels
 - B. Block ligand-gated Ca^{2+} channels
 - C. Block voltage-gated Na^+ channels
 - D. Stimulate ligand-gated Na^+ channels
 - E. Stimulate voltage-gated Ca^{2+} channels
26. A 40-year-old male presented with hand injury need immediate suture. Lidocaine is given as a local anesthesia. What is the mechanism of lidocaine? [NL 2021]
- A. Activation of muscarinic acetylcholine receptor
 - B. Activation of ligand-gated potassium channel
 - C. Inhibition of nicotinic acetylcholine receptor
 - D. Inhibition of voltage-gated sodium channel
 - E. Inhibition of acetylcholine release
27. Muscle weakness in the afternoon and evening. No symptoms in the morning. What is the pathogenesis of this disease [NL 2024]?
- A. Autoantibody to Ach receptor
 - B. Deficiency of acetylcholinesterase
 - C. Voltage-gated calcium channel antibodies
 - D. Mutation in the SCN4A gene
 - E. Impaired calcium release from the sarcoplasmic reticulum
28. What is the major energy usage in the brain?
- A. Action potential
 - B. resting potential
 - C. Postsynaptic Ca^{2+}
 - D. Glutamate reuptake
 - E. Vesicle recycling
29. Use Na^+/K^+ pumping during?
- A. depolarization
 - B. repolarization
 - C. hyperpolarization
 - D. resting stage
 - E. Only during action potential firing, never at rest

- 30.** What is it called when depolarization from multiple synapses summates?
- A. Presynaptic facilitation
 - B. Presynaptic inhibition
 - C. spatial summation
 - D. Temporal summation
 - E. Long-term potentiation
- 31.** All of the following are correct about chemical synapse EXCEPT which one?
- A. Action potential at presynaptic terminal open Ca^{2+} channel
 - B. NT can stimulate ligand gated Na^{+} channel
 - C. No synaptic delay
 - D. NT reuptake and broken down by enzyme
 - E. Postsynaptic receptors can be either ionotropic or metabotropic
- 32.** Which statement about graded potentials is incorrect?
- A. Their magnitude varies directly with the strength of the stimulus.
 - B. They can involve either depolarization or hyperpolarization.
 - C. They are short-lived, local changes in membrane potential.
 - D. They transmit signals over long distances without decreasing in strength.
 - E. They decrease in intensity with distance.

Lecture 4 — Metabolic Pathways of Neurons (5 questions)

- 33.** Which of the followings is INCORRECT?
- A. glutamate reuptake to astrocyte require ATP
 - B. ATP production in astrocyte mostly come from glycolysis
 - C. ATP production in neuron mostly come from oxidative phosphorylation
 - D. Astrocyte can convert reduced ascorbic acid to oxidized ascorbic acid
 - E. Monocarboxylate transporter (MCT) is lactate transporter
- 34.** Which of the followings is INCORRECT?
- A. Monocarboxylate transporter (MCT) is lactate transporter.
 - B. ATP production in astrocyte mostly come from glycolysis.
 - C. α -ketoglutarate in TCA cycle is a precursor of glutamate.
 - D. Glycogen synthase is the most active enzyme and found in neuron.
 - E. Neurons rely heavily on oxidative phosphorylation for ATP production
- 35.** What enzyme hydrolyzes glutamine to glutamate? (MD27,28)
- A. Glutaminase
 - B. Glutamine synthetase
 - C. Glutamate decarboxylase
 - D. Glycogen synthase
 - E. Monoamine oxidase

36. Accumulation of glutamate causes toxicity due to the accumulation of what ion in the cell?

- A. Calcium
- B. Sodium
- C. Glutamine
- D. Glutamate
- E. Chloride

37. What enzyme hydrolyzes glutamine to glutamate?

- A. Glutaminase
- B. Glutamine synthetase
- C. Glutamate decarboxylase
- D. Glycogen synthase
- E. Monoamine oxidase

Lecture 5 — Embryonic Development of the NS (12 questions)

38. Which of the following structures developed to be substantia nigra?

- A. Metencephalon
- B. Myelencephalon
- C. Telencephalon
- D. Diencephalon
- E. Mesencephalon

39. The optic cup is an evagination of which of the following?

- A. Telencephalon
- B. Diencephalon
- C. Mesencephalon
- D. Metencephalon
- E. Myelencephalon

40. Which of the following diseases have the characteristic “protrusion of meninges”?

- A. Anencephaly
- B. Meroencephaly
- C. Spina bifida occulta
- D. Spina bifida with meningocele
- E. Spina bifida with meningomyelocele

41. Which of the following structures developed to the pineal gland?

- A. Diencephalon
- B. Telencephalon
- C. Metencephalon
- D. Mesencephalon
- E. Myelencephalon

42. Which of the following represents the general somatic efferent (GSE) at the myelencephalon?
- A. Trochlear nucleus
 - B. Abducens nucleus
 - C. Hypoglossal nucleus
 - D. Dorsal motor nucleus of vagus
 - E. Nucleus ambiguus
43. Which of the following structures differentiates into the optic cup?
- A. Endoderm
 - B. Mesoderm
 - C. Neuroectoderm
 - D. Neural crest cells
 - E. Surface ectoderm
44. What birth defect of the spinal cord is associated with hydrocephalus?
- A. meningocele
 - B. meningocele
 - C. myeloschisis
 - D. spina bifida occulta
 - E. Anencephaly
45. Which condition results from the failure of neural folds to fuse?
- A. meningocele
 - B. meningocele
 - C. myeloschisis
 - D. spina bifida occulta
 - E. Encephalocele
46. What is the developmental origin of the pituitary gland?
- A. Diencephalon
 - B. Metencephalon
 - C. Myelencephalon
 - D. Telencephalon
 - E. Neuroectoderm generally, unspecified
47. What does corneal endothelium develop from?
- A. surface ectoderm
 - B. mesoderm
 - C. neural crest cells
 - D. neuroectoderm
 - E. Endoderm
48. What is the condition from failure in Anterior neuropore closure?
- A. Meningoencephalocele
 - B. Meroencephaly
 - C. Omphalocele
 - D. Ventricular septal defect
 - E. Spina bifida occulta

49. What does the lens develop from?

- A. Surface ectoderm
- B. Neuroectoderm
- C. Mesoderm
- D. Neural crest
- E. Endoderm

Lecture 6 — Development of the Eye (4 questions)

50. High intracerebral pressure resulting in which of the following signs and symptoms?

- A. Papillary edema
- B. Compression of optic cup
- C. Macular degeneration
- D. Retinal detachment
- E. Optic atrophy

51. What is derived from the ventral part of the optic vesicle?

- A. Choroid fissure
- B. Retinal fissure
- C. Optic stalk lumen
- D. Hyaloid canal
- E. Lens vesicle remnant

52. Failure of the choroid fissure to fuse results in?

- A. Coloboma
- B. Anophthalmia
- C. Color Blindness
- D. Retina dysfunction
- E. Cyclopia

53. The spinal trigeminal nucleus is derived from what?

- A. Basilar part
- B. Alar part
- C. Surface ectoderm
- D. Neural crest cells
- E. Notochord

Lecture 7 — Spinal Cord & Blood Supply (16 questions)

54. The result of the MRI exhibits hemisection of spinal cord (Brown sequard syndrome) at the right L2 spinal nerve level. Which of the following sensations losses on the same side at the right L2 level and below?

- A. Temperature
- B. Pain
- C. Conscious proprioception
- D. Crude touch
- E. Motor function only, no sensory involvement

55. A 65-year-old man has lost vibration, position and joint discrimination in the legs and lower body. Lesions to which of the following areas of the spinal cord are most likely to be seen?

- A. Ventral column (funiculus)
- B. Dorsal column (funiculus)
- C. Lateral column (funiculus)
- D. Dorsal (posterior) horn
- E. Intermediate zone of the gray matter

56. Stiff neck is a compensated sign to limit head and neck movements of a patient who is likely affected at which of the following structures?

- A. Brain
- B. Spinal cord
- C. Choroid plexus
- D. Meningeal dura
- E. Cerebral blood vessels

57. A patient has an injury that damages the lateral horn of the spinal cord. Which of the following would you expect to see in the patient?

- A. Loss of crude touch
- B. Loss of proprioceptive
- C. Loss of sympathetic impulse
- D. Loss of pain and temperature
- E. Loss of the voluntary motor impulse

58. All of the following statements concerning the anterior spinal artery are correct EXCEPT

- A. A branch of the vertebral artery
- B. Supplied anterior 2/3 of spinal cord
- C. Within the anterior median fissure
- D. Runs with the anterior spinal vein
- E. Occlusion would cause loss of proprioception

59. Stiff neck is a compensated sign to limit head and neck movements of a patient who is likely irritated at which of the following structures?

- A. Brain
- B. Spinal cord
- C. Choroid plexus
- D. Meningeal dura
- E. Cerebral blood vessels

60. A 53-year-old male was hospitalized for an acute spinal cord infarction. He presented with bilateral motor dysfunction, loss of pain and temperature below infarct levels, and preservation of proprioception and vibratory sense. Which of the following is the most likely diagnosis?

- A. Complete cord syndrome
- B. Brown-Sequard syndrome
- C. Anterior spinal cord syndrome
- D. Central cord syndrome
- E. Posterior spinal cord syndrome

- 61.** The result of MRI exhibits hemisection of the spinal cord (Brown Sequard syndrome) at the Right L2 spinal cord level. Which of the following is correct about the Rt. Brown Sequard syndrome?
- A. Contralateral loss of fine touch from L2 and below
 - B. Ipsilateral loss of pain and temperature below the L1 segment
 - C. Contralateral loss of two-point discrimination from L2 and below
 - D. Contralateral loss of pain and temperature from the L4 and below
 - E. Bilateral loss of all sensory modalities
- 62.** A 55-year-old man suffered an injury that caused damage to the dorsal column of his spinal cord. Which of the following signs would you expect to find in the patient?
- A. Loss of touch
 - B. Loss of proprioceptive
 - C. Loss of pain and temperature
 - D. Loss of voluntary motor impulse
 - E. Loss of sympathetic impulse
- 63.** Stiff neck is a compensated sign to limit head and neck movements of a patient who is likely irritated at which of the following structures?
- A. Brain
 - B. Spinal cord
 - C. Choroid plexus
 - D. Meningeal dura
 - E. Cerebral blood vessels
- 64.** What function is lost with damage to the lateral horn of the spinal cord?
- A. Intermediolateral nucleus
 - B. Sacral parasympathetic nucleus
 - C. Intermediomedial nucleus
 - D. Nucleus proprius
 - E. Substantia gelatinosa
- 65.** What is a likely consequence of posterior spinal cord artery damage?
- A. leg weakness
 - B. loss pain sensation
 - C. loss of vibration and proprioception
 - D. Loss of motor function
 - E. Loss of temperature sensation
- 66.** Which of the following is not part of the CNS?
- A. hypothalamus
 - B. spinal cord
 - C. basal ganglion
 - D. vagal ganglion
 - E. Thalamus

67. If the anterior spinal artery is damaged, what is spared?
- A. Pain
 - B. temperature
 - C. crude touch
 - D. Motor function
 - E. Proprioception
68. Which of the following is a primary motor symptom associated with a complete spinal cord lesion?
- A. Hemiparesis
 - B. Isolated sensory loss
 - C. Tetraplegia or Paraplegia
 - D. Impaired fine motor control only
 - E. Unilateral reflex absence
69. What extends from the pia mater to the dura mater, and separates the dorsal and ventral nerve roots?
- A. Conus medullaris
 - B. filum terminale
 - C. coccygeal ligament
 - D. denticulate ligament
 - E. Arachnoid trabeculae

Lecture 8 — PNS: Cranial & Spinal Nerves (16 questions)

70. Which of the functional components below is found in the hypoglossal nucleus?
- A. GSA
 - B. SVA
 - C. GVA
 - D. SVE
 - E. GSE
71. According to the Peripheral Nervous System (PNS), which of the statements below is INCORRECT?
- A. Spinal nerve is formed by a union of anterior and posterior nerve roots, and passes through the intervertebral foramen.
 - B. Trigeminal ganglion contains pseudounipolar neurons that receive somatic sense from skin of face, eye, nasal cavity meninges and anterior 2/3 of tongue.
 - C. There is only 1 functional component found in the accessory nerve.
 - D. Secretomotor functions from brainstem are transmitted via the CN. III, VII, IX, X.
 - E. There are 12 pairs of cranial nerves, and only CN. III - XII emerge from the brainstem.
72. Collections of nerve cell bodies inside the peripheral nervous system are called:
- A. Tracts
 - B. nerve fibers
 - C. ganglia
 - D. nerves
 - E. Nuclei

73. Which of the functional components IS NOT found in the facial nerve?

- A. GSA
- B. GVE
- C. GSE
- D. SVA
- E. SVE

74. A 41-year-old woman complains to her family physician about recurring episodes of sharp pain that seem to originate from around her mouth and cheek. The pain is so intense that she is unable to eat, brush her teeth, or apply make-up. Which of the following cranial nerves is the most likely source of this pain?

- A. Facial (VII)
- B. Glossopharyngeal (IX)
- C. Hypoglossal (XII)
- D. Trigeminal (V)
- E. Vagus (X)

75. Which of the following spinal nerves are NOT included in a nerve plexus?

- A. C1-C5
- B. C5-T1
- C. T2-T12
- D. L1-L4
- E. L4-S4

76. When the light gets into the eye, which structures absorb the light?

- A. Choroid
- B. Ciliary body
- C. Outer nuclear layer
- D. Inner nuclear layer
- E. Ganglion cell layer

77. Which of the following spinal nerves are NOT included in a nerve plexus?

- A. C1-C5
- B. C5-T1
- C. T2-T12
- D. L1-L4
- E. L4-S4

78. Which of the functional component of nerve fibers IS NOT found in a spinal nerve?

- A. GSA
- B. GSE
- C. GVA
- D. GVE
- E. SVE

79. Collections of nerve cell bodies (soma) inside the peripheral nervous system are called

- A. Tracts
- B. Nuclei
- C. Nerves
- D. Ganglia
- E. Rootlets

80. Which of the functional components IS NOT found in the facial nerve?
- A. GSA
 - B. GVE
 - C. GSE
 - D. SVA
 - E. SVE
81. Which nucleus contains General Visceral Efferent (GVE) fibers?
- A. inferior olivary nu.
 - B. Dorsal motor nucleus of Vagus
 - C. Inferior Vestibular nu.
 - D. inferior salivatory nu.
 - E. Nucleus ambiguus
82. Which of the following is correct about the transverse section of caudal pons?
- A. pontine nuclei lies between transverse pontine fiber
 - B. vestibular nuclei medial abducens nucleus
 - C. MLF is in the tectum above the fourth ventricle
 - D. Facial nucleus is located medially in the caudal pons
 - E. Trigeminal motor nucleus is in the caudal pons
83. An optical disc cannot detect light energy because of the absence of?
- A. Absent blood vessel
 - B. Absent nerve
 - C. Absent of ganglion cell
 - D. Absent Photoreceptor
 - E. Absent pigment epithelium
84. What dermatome is listed incorrectly?
- A. Shoulder – C5
 - B. Posterior of thigh and leg – T1-2
 - C. Umbilical – T10
 - D. Little finger – C8
 - E. Nipple line – T4
85. Which nerve fiber type is responsible for increased salivation & constriction of the urinary bladder?
- A. General Somatic Afferent (GSA)
 - B. General Somatic Efferent (GSE)
 - C. Special Visceral Efferent (SVE)
 - D. Special Somatic Afferent (SSA)
 - E. General Visceral Efferent (GVE)

Lecture 9 — Brainstem Structures (17 questions)

- 86.** The MRI result appears compression of a tumor at the right tuberculum cinerium of close medulla. Which of the following sensations is disturbed in this pathology?
- A. Vibration sense of right side of the body
 - B. Pain and temperature of the left side of the body
 - C. Pain and temperature of the right face
 - D. Proprioception of right side of the body
 - E. Motor weakness of right face
- 87.** The MRI result appears compression of a tumor at Right Tuberculum cinerium of close medulla. Which of the following structures formed the 'tuberculum cinerium'?
- A. Inferior olive
 - B. Spinal trigeminal nucleus and tract
 - C. Gracile nucleus and fasciculus gracilis
 - D. Cuneate nucleus and fasciculus cuneatus
 - E. Nucleus ambiguus
- 88.** Which of the following reflex tests represents functions of the medulla oblongata?
- A. Gag reflex
 - B. Corneal reflex
 - C. Jaw jerk reflex
 - D. Pupillary light reflex
 - E. Doll's eyes (Oculocephalic) reflex
- 89.** A 20-year-old man w/ loss of muscle power at the left side of the body and his right eye has strabismus (). Pupillary light reflex show pupil dilated (mydriasis) in the right side. What specific level and side of the brainstem is most likely damaged?
- A. Mid pons, right side
 - B. Closed medulla, left side
 - C. Rostral midbrain, right side
 - D. Caudal pons, left side
 - E. Caudal medulla, right side
- 90.** The MRI result appears compression of a tumor at Right tuberculum cinereum of close medulla. Which of the following nerves does not give the axons to synapse at the compression of tuberculum cinereum?
- A. Trigeminal nerve
 - B. Facial nerve
 - C. Glossopharyngeal nerve
 - D. Vagus nerve
 - E. Accessory nerve
- 91.** Which of the reflex tests represents functions of the medulla oblongata?
- A. Gag reflex
 - B. Corneal reflex
 - C. Jaw jerk reflex
 - D. Pupillary light reflex
 - E. Doll's eyes (Oculocephalic) reflex

92. Which of the following neuronal pathways performs voluntary control of body and limb movements?

- A. Reticulospinal tract
- B. Corticopontine tract
- C. Posterior column and medial lemniscus pathway
- D. Hypothalamospinal tract
- E. Pyramidal tract

93. Which of the following CNS structures locates at the same horizontal level of tentorial cere- belli?

- A. Pons
- B. Midbrain
- C. Thalamus
- D. Cerebellum
- E. Medulla oblongata

94. A 32-year-old man loss of muscle power, presented Babinski sign at the left side. His right eye has lateral strabismus () and pupillary light reflex show Rt. pupil dilate (mydriasis). What specific level and side of the brainstem is most likely damaged?

- A. Mid pons, right side
- B. Closed medulla, left side
- C. Rostral midbrain, right side
- D. Caudal pons, left side
- E. Caudal medulla, right side

95. MRI result appears compression of a tumor at Right tuberculum cinereum of close medulla. Which of the following nerves DOES NOT give the axons to synapse at the compression of tuberculum cinereum?

- A. Vagus
- B. Trigeminal
- C. Accessory
- D. Glossopharyngeal
- E. Facial

96. Which of the reflex tests represents functions of the medulla oblongata?

- A. Gag reflex
- B. Corneal reflex
- C. Jaw jerk reflex
- D. Pupillary light reflex
- E. Doll's eyes (Oculocephalic) reflex

97. A patient presents with left eye symptoms where the pupil is larger than the right, and a history of lower facial weakness. Where is the lesion located?

- A. Midbrain
- B. Pons
- C. Medulla
- D. Cerebral cortex
- E. Spinal cord

- 98.** A patient is diagnosed with loss of pain and temperature contralateral from the body and ipsilateral from the face. Where is the site of lesion?
- A. Medial medulla oblongata
 - B. Lateral medulla oblongata
 - C. Caudal pons
 - D. Rostral pons
 - E. Midbrain
- 99.** Which tract's secondary neuron synapses and crosses to the opposite side at the medulla oblongata?
- A. anterior corticospinal tract
 - B. lateral spinothalamic tract
 - C. Lateral corticospinal tract
 - D. Fasciculus gracilis
 - E. Anterior spinothalamic tract
- 100.** Damage to which major component of the Ascending Reticular Activating System (ARAS) would most directly affect consciousness?
- A. Cerebellum
 - B. Thalamus
 - C. Hypothalamus
 - D. Pons and Midbrain
 - E. Cerebrum
- 101.** What is the major site of the ARAS related to coma?
- A. Hypothalamus
 - B. Cerebral neocortex
 - C. Pons and midbrain
 - D. Thalamus
 - E. Basal ganglia
- 102.** A male patient presents with alteration of consciousness, irregular breathing, and abnormal limb posturing. Lesion in which of the following can lead to his presentation?
- A. Cerebrum
 - B. Midbrain
 - C. Pons
 - D. Medulla
 - E. Cerebellum

Lecture 10 — Reticular Formation (3 questions)

- 103.** Which of the following pairs is the correct match (related functions)?
- A. Fasciculus gracilis – parasympathetic nucleus
 - B. Lateral spinothalamic tract – substantia gelatinosa
 - C. Dorsal spinocerebellar tract - nucleus proprius
 - D. Lateral corticospinal tract - Clarke's nucleus
 - E. Ventral spinocerebellar tract - inferior olivary nucleus

110. Which one affects the PNS?

- A. Alzheimer
- B. Brown sequard syndrome
- C. Carpal tunnel syndrome
- D. Multiple sclerosis
- E. Parkinson's disease

111. A patient has right-sided pain and temperature loss from T12, left-sided pain and temperature loss from T10, left-sided proprioception loss from T10, and left-sided Horner syndrome. What is the condition?

- A. central cord syndrome at T10
- B. brown sequard at T10
- C. brown sequard at T12
- D. at T12
- E. Anterior cord syndrome at T10

112. What structure for unconscious proprioception at S2-4 synapses at Clarke's nucleus?

- A. Dorsal spinocerebellar
- B. Cuneate funiculus
- C. Gracile funiculus
- D. spinothalamic tract
- E. Lateral corticospinal tract

113. Which unconscious tract synapses with Clarke's nucleus?

- A. ventral spinocerebellar
- B. lateral spinothalamic
- C. Corticospinal tract
- D. dorsal spinocerebellar
- E. dorsal column

Lecture 13 — Taste (Gustatory) Sensation (6 questions)

114. According to the lesion in a providing picture, what are the expected signs and symptoms of this lesion?

- A. Tongue deviates to the left side, Right hemiplegia, Loss of light touch and proprioception senses at the left side
- B. Tongue deviates to the right side, Left hemiplegia, Loss of light touch and proprioception senses at the left side
- C. Tongue deviates to the right side, Left hemiplegia, Loss of light touch and proprioception senses at the right side
- D. Tongue deviates to the right side, Right hemiplegia, Loss of light touch and proprioception senses at the right side
- E. Tongue deviates to the left side, Right hemiplegia, Loss of light touch and proprioception senses at right side

115. Autonomic controlling of the heart, lungs, digestion, taste, communication between brain and organs is.....
nerves

- A. Hypoglossal
- B. Vagus
- C. Vestibulocochlear
- D. Trigeminal
- E. Abducens

116. A 71-year-old man complains to his family physician that his face 'feels funny.' The examination reveals numbness on his face and on the same side of his tongue. MRI shows a lesion to the cerebral cortex. This man's lesion is most likely located in which of the following cortical regions?

- A. Lateral one-third of the postcentral
- B. Anterior paracentral
- C. Lateral one-third of the precentral
- D. Middle one-third of the postcentral
- E. Posterior paracentral

117. According to the lesion in a providing picture, what are the expected signs and symptoms of this lesion?

- A. Tongue deviates to the left side, Right hemiplegia, Loss of light touch and proprioception senses from right side.
- B. Tongue deviates to the left side, Right hemiplegia, Loss of light touch and proprioception senses from the left side.
- C. Tongue deviates to the right side, Left hemiplegia, Loss of light touch and proprioception senses from the left side.
- D. Tongue deviates to the right side, Right hemiplegia, Loss of light touch and proprioception senses from the right side.
- E. Tongue deviates to the right side, Left hemiplegia, Loss of light touch and proprioception senses from right side.

118. Autonomic controlling of the heart, lungs, digestion and taste communication between brain and organs is..... nerve.

- A. Vagus
- B. Hypoglossal
- C. Trigeminal
- D. Abducens
- E. Vestibulocochlear

119. What is the correct pathway for taste sensation from the anterior 2/3 of the tongue?

- A. Facial nerve (Chorda tympani) → Geniculate ganglion → Solitary nucleus (upper part) → VPM of thalamus → Gustatory area of cerebral cortex.
- B. Glossopharyngeal nerve → Petrosal ganglion → Solitary nucleus → VPL of thalamus → Postcentral gyrus.
- C. Vagus nerve → Inferior (nodose) ganglion → Solitary nucleus → Medial geniculate body → Primary auditory area.
- D. Trigeminal nerve → Trigeminal ganglion → Spinal trigeminal nucleus → VPM of thalamus → Primary somesthetic area.
- E. Facial nerve → Cochlear nucleus → Superior olivary complex → Inferior colliculus → Primary auditory cortex.

Lecture 14 — Sensory Receptors & Neurotransmitters (35 questions)

120. A woman presented with depressive symptoms and suicidal thoughts. Which of the following neurotransmitters is abnormal in this case? [NL 2016]

- A. Glutamate
- B. Acetylcholine
- C. Serotonin
- D. GABA
- E. Dopamine

121. What is the mechanism of action of botulinum toxin? [NL 2017]

- A. Increased sodium channel at presynaptic neuron
- B. Blocks acetylcholine receptor at postsynaptic neuron
- C. Blocks presynaptic fusion of neurotransmitter vesicle
- D. Blocks calcium channel at presynaptic neuron
- E. Increased reabsorption of acetylcholine

122. An 80-year-old man had a history of progressive short-term memory loss and problems with words in speaking and writing. He was diagnosed with Alzheimer disease. Which of the following about pathogenesis of Alzheimer disease is INCORRECT?

- A. Atrophy of gyri in cerebral cortex
- B. The formation of amyloid plaques
- C. The formation of neurofibrillary tangles
- D. Increase activity of beta-secretase enzyme
- E. Blocking transporters for dopamine reuptake

123. A patient came to the hospital with continuous generalized tonic-clonic seizures for 15 minutes. What is the most appropriate first line treatment? [NL 2016]

- A. Diazepam
- B. Valproate
- C. Phenytoin
- D. Fluoxetine
- E. Memantine

124. The summation of EPSPs from numerous presynaptic nerve fibers converging onto one postsynaptic neuron is called. . .

- A. Spatial summation
- B. Temporal summation
- C. Presynaptic inhibition
- D. Long-term potentiation
- E. Presynaptic facilitation

125. A 20-year-old college student presents with depressive symptoms, including sleep disturbance and anorexia for 1 month. This patient has never had any mood disorders. What is the mechanism of the drug used to treat this patient? [NL 2020, 2017]

- A. Stimulation of GABA receptors
- B. Inhibition of dopamine receptors
- C. Inhibition of serotonin transporter
- D. Inhibition of acetylcholinesterase
- E. Stimulation of norepinephrine release

126. A woman presents with progressive muscle weakness. PE: dysphagia, ptosis when asked to perform repeated blinking. Which of the following is most likely present in this case? [NL 2016, 2017]

- A. Ryanodine receptor antibody
- B. Anti-presynaptic Ca^{2+} antibody
- C. Anti-acetylcholine receptor antibody
- D. Inhibition of dopamine receptors
- E. Stimulation of GABA receptors

- 127.** A 70-year-old patient presents with dementia and cognitive problems. He received donepezil as a treatment. Which of the following neurotransmitters is affected as a result of this drug? [NL2019]
- A. Norepinephrine
 - B. GABA
 - C. Glutamate
 - D. Acetylcholine
 - E. 5-HT
- 128.** An 80-year-old man had history of progressive short-term memory loss and problems with words in speaking and writing. He was diagnosed with Alzheimer disease. Which of the following about pathogenesis of Alzheimer disease is INCORRECT?
- A. Atrophy of gyri in cerebral cortex
 - B. The formation of amyloid plaques
 - C. The formation of neurofibrillary tangles
 - D. Increase activity of β -secretase enzyme
 - E. Blocking transporters for dopamine reuptake
- 129.** A 30-year-old woman presents with bilateral ptosis and blurred vision. A complete history reveals that the woman has also been experiencing generalized fatigue and weakness. Which of the following is the cause of the disease in this patient? [NL2023]
- A. Increase dopamine.
 - B. Decrease acetylcholine.
 - C. Inhibition of acetylcholine receptor.
 - D. Increase activity of acetylcholinesterase.
 - E. Increased reabsorption of acetylcholine.
- 130.** A woman presented with depressive symptoms and suicidal thoughts. Which of the following neurotransmitter is abnormal in this case? [NL 2016]
- A. Serotonin
 - B. Dopamine
 - C. Acetylcholine
 - D. Glutamate
 - E. GABA
- 131.** A 30-year-old woman presents with bilateral ptosis and blurred vision. A complete history reveals that the woman has also been experiencing generalized fatigue and weakness. Which of the following is the cause of the disease in this patient? [NL2023]
- A. Increase dopamine.
 - B. Decrease acetylcholine.
 - C. Inhibition of acetylcholine receptor.
 - D. Increase activity of acetylcholinesterase.
 - E. Increased reabsorption of acetylcholine.
- 132.** What is the mechanism of action of botulinum toxin?
- A. Blocks presynaptic fusion of neurotransmitter vesicle
 - B. Blocks calcium channel at presynaptic neuron
 - C. Blocks acetylcholine receptor at postsynaptic neuron
 - D. Increased reabsorption of acetylcholine
 - E. Blocks postsynaptic K^+ channels

133. A 45-year-old male presents to the emergency department with a deep wound on his foot following a gardening accident. He reports increasing stiffness and difficulty opening his jaw. On examination, his jaw is found to be locked, and he exhibits muscle spasms. Which of the following mechanisms best describes the action of the toxin responsible for the patient's symptoms?

- A. Inhibit neurotransmitter release
- B. Autoantibody to nicotinic acetylcholine receptors (nAChR)
- C. Inhibit calcium channels at the postsynaptic membrane
- D. Block inhibitory interneurons
- E. Increases presynaptic Ca^{2+} influx

134. A 20-year-old college student presents with depressive symptoms, including sleep disturbance and anorexia for 1 month. This patient has never had any mood disorders. What is the mechanism of the drug used to treat this patient?

- A. Stimulation of GABA receptors
- B. Inhibition of dopamine receptors
- C. Inhibition of serotonin transporter
- D. Stimulation of norepinephrine release
- E. Inhibition of MAO-A

135. A woman presents with auditory hallucination and delusion. Which of the following neurotransmitters is most likely associated with these symptoms?

- A. Adrenaline
- B. Gamma-aminobutyric acid
- C. Dopamine
- D. Norepinephrine
- E. Serotonin

136. A 70-year-old patient presents with dementia and cognitive problems. He received donepezil as a treatment. Which of the following neurotransmitters is affected as a result of this drug?

- A. GABA
- B. Glutamate
- C. Acetylcholine
- D. 5-HT
- E. Dopamine

137. An 80-year-old man had history of progressive short-term memory loss and problems with words in speaking and writing. He was diagnosed with Alzheimer disease. Which of the following about pathogenesis of Alzheimer disease is INCORRECT?

- A. Atrophy of gyri in cerebral cortex
- B. The formation of amyloid plaques
- C. The formation of neurofibrillary tangles
- D. Increase activity of β -secretase enzyme
- E. Blocking transporters for dopamine reuptake

- 138.** A patient receiving chemotherapy experiencing nausea and vomiting which antiemetic medication should be administered that inhibits what? [NL 2024]
- A. 5-HT₃ receptor
 - B. Dopamine receptor
 - C. Muscarinic receptor
 - D. Nicotinic receptor
 - E. Adrenergic receptor
- 139.** Which stimuli (agents) do not stimulate the nociceptor?
- A. Prostaglandin
 - B. K⁺
 - C. Substance P
 - D. Histamine
 - E. Vibration
- 140.** What is the term for the time between the arrival of an action potential at a synaptic terminal and its effect on the postsynaptic membrane?
- A. synaptic delay
 - B. synaptic fatigue
 - C. Refractory period
 - D. Threshold potential
 - E. Conduction velocity
- 141.** Which receptor is ionotropic?
- A. AMPA
 - B. mGluR6
 - C. GABA_b
 - D. 5-HT₁
 - E. Muscarinic ACh receptor
- 142.** What is the mechanism of fluoxetine?
- A. selective serotonin uptake inhibitors
 - B. block D₂ receptor
 - C. block peripheral and central 5-HT₃ receptor
 - D. Inhibit MAO
 - E. Block NMDA receptor
- 143.** What is the mechanism of the drug ondansetron?
- A. Selective serotonin uptake inhibitors
 - B. Block D₂ receptor
 - C. Block peripheral and central 5-HT₃ receptor
 - D. Block muscarinic receptor
 - E. Stimulate dopamine receptor
- 144.** What is the mechanism of haloperidol?
- A. Selective serotonin uptake inhibitors
 - B. Block D₂ receptor
 - C. Block peripheral and central 5-HT₃ receptor
 - D. Block NMDA receptor
 - E. Inhibit acetylcholinesterase

- 145.** A patient got splashed by heated oil during cooking. Which of the following is the receptor (channel) for heat in this case?
- A. TRPV2
 - B. TRPV4
 - C. TRPM8
 - D. TRPA1
 - E. TRPV1
- 146.** A woman with fatigue, social withdrawal, and depression likely has an issue with which neurotransmitter?
- A. Dopamine
 - B. Glutamate
 - C. Serotonin
 - D. Acetylcholine
 - E. Epinephrine
- 147.** What does Fluoxetine inhibit?
- A. SERT
 - B. EAATs
 - C. DAT
 - D. VMAT
 - E. NET
- 148.** A 75-year-old woman with progressive memory loss and cognitive decline received memantine and galantamine as treatment. Which neurotransmitter is affected by Galantamine?
- A. Serotonin
 - B. Acetylcholine
 - C. GABA
 - D. Dopamine
 - E. Glutamate
- 149.** Which of these is not an ionotropic receptor?
- A. AMPA receptor
 - B. Nicotinic Acetylcholine Receptor (nAChR)
 - C. GABA B Receptor
 - D. 5-HT₃ receptor
 - E. Glycine receptor
- 150.** All of the following are G-protein-coupled receptors EXCEPT?
- A. Dopamine D₂ receptor
 - B. Muscarinic acetylcholine receptor
 - C. mGluR6 receptor
 - D. 5-HT₁ receptor
 - E. 5-HT₃ of Serotonin
- 151.** A patient is treated with Donepezil and Memantine for Alzheimer's disease. What is the effect of Memantine?
- A. Acetylcholinesterase inhibition
 - B. Dopamine D₂ receptor blockade
 - C. Glutamate (NMDA receptor antagonist)
 - D. Serotonin transporter inhibition
 - E. GABA receptor stimulation

152. Which receptor is primarily blocked by haloperidol?

- A. Dopamine D2 receptor
- B. 5-HT3 receptor
- C. Muscarinic acetylcholine receptor
- D. GABA receptor
- E. NMDA receptor

153. Loss of smell and taste (anosmia and ageusia) can result from a deficiency in which receptor, also known for its role in viral entry?

- A. Dopamine D2 receptor
- B. Muscarinic acetylcholine receptor
- C. ACE2
- D. 5-HT3 receptor
- E. NMDA receptor

154. Which of the following neurotransmitter-transporter pairs is correct?

- A. VGAT - GABA
- B. SERT - Dopamine
- C. DAT - Acetylcholine
- D. EAATs - Norepinephrine
- E. Glutaminase - Glutamine

Lecture 15 — Visual System (27 questions)

155. A patient exhibits the near vision reflex defect. Which of the following DOES NOT the characteristic of near vision reflex?

- A. Medial deviation of both eyes
- B. Levator palpebrae superioris contraction
- C. Smaller diameter of pupil
- D. More convexity of lens
- E. Ciliary muscle contraction

156. A patient exhibits the near vision reflex defect. Which of the following nerves involves the near vision reflex?

- A. CN. III and parasympathetic nervous system
- B. CN. III and sympathetic nervous system
- C. CN. IV and CN. VI and sympathetic nervous system
- D. CN. VI and parasympathetic nervous system E.
- E. CN II and CN III only, no ANS involvement

157. A patient has a tumor compression at the optic chiasma. Which of the signs and symptoms would be found in this patient?

- A. Left homonymous hemianopia.
- B. Right homonymous hemianopia.
- C. Bitemporal heteronymous hemianopia
- D. binasal heteronymous hemianopia
- E. Complete bilateral blindness

158. Which of the following nerve fibers involves this near vision reflex?

- A. Abducent nerve
- B. Trochlear nerve
- C. Parasympathetic nerve fiber
- D. Sympathetic nerve fiber
- E. Facial nerve

159. Which of the following structures would likely damage after examination the light reflex and the result as follow the table below?

- A. Rt. CN. III
- B. Rt. optic nerve
- C. Lt. CN. III
- D. Lt. optic nerve
- E. Optic chiasma

160. Which of the following statements is true regarding visual systems?

- A. The monocular field of either eye presents 3 dimensions.
- B. Blind spot lies about 3- or 4-mm temporal to the fovea centralist.
- C. Object at the upper temporal quadrant of the visual field project to the lower temporal retinal field.
- D. Information from the right sides of visual fields finally synapses with the left primary visual area.
- E. The second-order neuron of the visual pathway is located at the brainstem.

161. Which part of the retina has no photoreceptive cells?

- A. Macula lutea
- B. Fovea centralis
- C. Retina that close to ora serrata
- D. Optic disc
- E. Paramacular field

162. When light shines into the right eye, the right pupil has no response, but the left pupil constricts. When light shines into the left eye, the right pupil has no response, but the left pupil is still contracted. Where is the lesion of the following structures?

- A. Right optic nerve
- B. Left optic nerve
- C. Right oculomotor nerve
- D. Left oculomotor nerve
- E. Right pretectal nucleus

163. What structure in the eye absorbs light?

- A. Choroid
- B. Retina
- C. Cornea
- D. Sclera
- E. Vitreous humor

164. What condition is indicated by an Intraocular Pressure (IOP) > 25mmHg?

- A. Cataract
- B. Retinal detachment
- C. Macular degeneration
- D. Glaucoma
- E. Uveitis

165. Which of the following correctly arranges the layers of the retina from the Rod and Cone cell layer to the optic nerve fiber layer?

- A. Rods and cones layer → Inner nuclear layer → Outer nuclear layer → Ganglion cell layer → Optic nerve fiber layer
- B. Rods and cones layer → Outer nuclear layer → Inner plexiform layer → Inner nuclear layer → Optic nerve fiber layer
- C. Rods and cones layer → Outer nuclear layer → Inner nuclear layer → Ganglion cell layer → Optic nerve fiber layer
- D. Rods and cones layer → Ganglion cell layer → Inner nuclear layer → Optic nerve fiber layer
- E. Rods and cones → Ganglion cell layer → Outer nuclear layer → Optic nerve fiber layer

166. What is the result of damage to the primary somesthetic sensory area?

- A. Motor paralysis
- B. Visual field defects
- C. Auditory hallucinations
- D. Impaired fine motor control
- E. Anesthesia

167. A patient presents with a left pupil that does not constrict and is larger than the right pupil, with a history of lower facial weakness. Where is the lesion located?

- A. optic tract
- B. optic nerve
- C. optic chiasm
- D. CN III
- E. optic radiation

168. Surface ectoderm gives rise to....?

- A. lens
- B. Iris
- C. Corneal endothelium
- D. Fibrous tissue of eye
- E. Retina

169. Which of the following specializations of the fovea increases visual acuity?

- A. abundant of blood vessels here
- B. high ratio of photoreceptor cell to ganglion cell
- C. lateral displacement of ganglion cell
- D. increase photoreceptor of rods cell
- E. Thicker retinal pigment epithelium at the fovea

170. Damage to which nerve affects the near vision reflex?

- A. CN.III
- B. CN.II
- C. CN.IV
- D. CN.VI
- E. CN VII

171. What structure is damaged in Right Homonymous Hemianopia?

- A. Left optic tract
- B. Optic chiasma
- C. Superior left geniculate body
- D. Inferior left geniculate body
- E. Right optic tract

172. Which of the following abnormal cells cause the most color blindness? [

- A. Abnormal rod cells
- B. Abnormal ganglion cells
- C. Abnormal cone cells
- D. Abnormal bipolar cells
- E. Abnormal pigment epithelium cells

173. What is the effect of sympathetic fibers on vision? [

- A. It causes the pupil to constrict.
- B. It increases the curvature (convexity) of the lens.
- C. It causes the pupil to dilate.
- D. It enhances the visual acuity in bright light.
- E. It reduces the sensitivity of rod cells.

174. What is not correct about the near vision reflex?

- A. Medial deviation of both eyes
- B. Levator palpebrae superioris contraction
- C. smaller diameter of pupil
- D. More convexity of lens
- E. Ciliary muscle contraction

175. What is the effect of retinal detachment?

- A. No blood supply to photoreceptor cells
- B. No blood supply to bipolar cells
- C. No blood supply to ganglion cells
- D. Increased intraocular pressure
- E. Immediate improvement in vision

176. Reflex test using wisp of cotton to touch sclera, with results as shown in the table. Where is the problem?

(Table: Stimulate RT eye -> both blink. Stimulate LT eye -> no blink.)

- A. Left Trigeminal
- B. Right Trigeminal
- C. Left facial nerve
- D. Right Oculomotor Nerve
- E. Left Optic Nerve

177. What is the visual field defect in a lesion of the right Meyer loop?

- A. Left homonymous superior quadrantanopia
- B. Right homonymous inferior quadrantanopia
- C. Right homonymous superior quadrantanopia
- D. Left homonymous inferior quadrantanopia
- E. Left homonymous hemianopia with macular sparing

178. Arrange the following events in order:

- A. Activation of Transducin
- B. Closure of Na channel
- C. Activation of phosphodiesterase
- D. Structural Change of the Retinal photopigment
- E. Decreased intracellular cGMP

179. A lesion in the right Meyer's loop (part of the optic radiation) causes what visual field defect?

- A. Left homonymous superior quadrantanopia
- B. Right homonymous inferior quadrantanopia
- C. Right homonymous superior quadrantanopia
- D. Left homonymous inferior quadrantanopia
- E. Left homonymous hemianopia with macular sparing

180. A patient with a right temporal lobe occlusion is likely to have which visual field defect?

- A. Right homonymous hemianopia
- B. Bitemporal heteronymous hemianopia
- C. Superior left quadrantanopia
- D. Left homonymous inferior quadrantanopia
- E. Left homonymous hemianopia

181. During a corneal reflex test, touching the right cornea causes no blink in either eye, while touching the left cornea causes both eyes to blink. Which nerve is damaged?

- A. Left facial nerve
- B. Right optic nerve
- C. Right trigeminal nerve
- D. Left trigeminal nerve
- E. Right oculomotor nerve

Lecture 16 — Auditory & Vestibular Systems (21 questions)

182. Which of the following items is found at the level of rostral pons?

- A. Cerebral aqueduct
- B. Abducens nucleus
- C. Area postrema
- D. Motor and Principal nuclei of CN. V
- E. Inferior vestibular nucleus

183. A patient has facial nerve damage resulting in hyperacusis (). Which of the following would likely be interrupted in the auditory system?

- A. Trapezoid body is interrupted.
- B. Superior olivary is interrupted.
- C. Tensor tympani reflex is interrupted.
- D. Stapedius reflex is interrupted.
- E. Lateral lemniscus is interrupted

184. Which of the following items is found at the level of rostral-mid pons?

- A. Facial motor nucleus
- B. Red nucleus
- C. Area prostroma
- D. Motor and Principle nuclei of CN. V
- E. Medial vestibular nucleus

185. Which of the following vestibular nuclei gives the axons to the oculomotor nucleus? (only 4 choices)

- A. Medial vestibular
- B. Lateral vestibular
- C. Superior vestibular
- D. Inferior vestibular
- E. None of these project to CN III

186. Which structures release sound waves from the inner ear?

- A. Eardrum
- B. Oval window
- C. Round window
- D. Helicotrema
- E. Scala vascularis

187. Which of the following statements is true regarding sound stimulation?

- A. Vibration within the endolymph of scala vestibuli is transmitted through the vestibular membrane.
- B. Outer hair cells vibrate and touch the tectorial membrane, causing an action potential carried to the cochlear nerve.
- C. Inner hair cells vibrate to stimulate the tectorial membrane and cause an action potential, which carries through the peripheral process of the spinal ganglion.
- D. The high frequency of sound waves stimulates the Organs of Corti at the apex of the cochlear duct.
- E. Low frequency sound stimulates the base of the cochlear duct

188. What structure is used to detect movement when using an elevator (vertical acceleration)?

- A. Sup semicircular
- B. Lat semicircular
- C. Post semicircular
- D. Sacculle
- E. Utricle

189. What structure is primarily used to detect the motion of tilting the head up (vertical linear movement)?

- A. Utricle
- B. Sacculle
- C. Horizontal semicircular canal
- D. Superior semicircular canal
- E. Cochlea

190. What structure is primarily used to detect the motion of nodding the head forward?

- A. Utricle
- B. Sacculle
- C. Horizontal semicircular canal
- D. Superior semicircular canal
- E. Cochlea

191. When you extend your head back (posterior), what receptor is stimulated?

- A. Sup semicircular
- B. Lat semicircular
- C. Post semicircular
- D. Sacculle
- E. Utricle

192. What is the location of the vomiting center?

- A. Vestibular nuclei
- B. Nu. Ambiguus
- C. Area postrema
- D. Superior and Inferior Salivatory
- E. Nucleus solitarius

193. What structure is used to reduce surrounding sound (by inhibiting outer hair cells) to better listen to people talk?

- A. Stapedius muscle
- B. Tectorial membrane
- C. Cochlear nerve
- D. Superior olivary complex
- E. Inferior colliculus

194. What is stimulated when walking forward?

- A. Lateral semicircular
- B. Anterior semicircular
- C. Posterior semicircular
- D. macula utricle
- E. macula sacculle

195. Trouble hearing low frequency suggests a possible occlusion where?

- A. Apex of Vestibular membrane
- B. Apex of Basilar membrane
- C. Base of cochlear duct
- D. Base of Vestibular membrane
- E. Central of Basilar membrane

196. Which pathway of hearing is correct?

- A. Cochlear nerve > Cochlear nu. > Sup. Olivary > Medial Geniculate body
- B. Cochlear nu. > Cochlear nerve > Sup. Olivary > Medial Geniculate body
- C. Cochlear nerve > Cochlear nu. > Sup. Olivary > Lateral Geniculate body
- D. Cochlear nu. > Cochlear nerve > Sup. Olivary > inferior colliculus > Lateral Geniculate body
- E. Cochlear nerve > Sup. Olivary > Cochlear nu. > Medial Geniculate body

197. An 80-year-old patient presents with presbycusis hearing loss (50-8,000HZ). Which of the following is the most likely damaged?

- A. Endolymph
- B. Outer hair cell
- C. Tectorial membrane
- D. Stria vascularis
- E. Inner hair cell

198. A man slipped and was about to fall. He extended his upper and lower extremity. Which structure involved this reflex?

- A. Lateral spinal vestibular tract
- B. Lateral vestibulospinal tract
- C. Medial vestibulospinal tract
- D. Medial spinal vestibular tract
- E. Cerebellospinal tract

199. What causes Presbycusis? [

- A. Damage to the tympanic membrane
- B. Stiffening of the ear ossicles
- C. Damage to hair cells
- D. Blockage of the Eustachian tube
- E. Excess fluid in the semicircular canals

200. What is incorrect about stimulating the crista ampullaris? [

- A. It detects angular acceleration of the head.
- B. Its hair cells are stimulated by the movement of endolymph.
- C. It contains sensory hair cells whose stereocilia are embedded in a cupola
- D. It detects linear acceleration of the head.
- E. Endolymph movement bends the stereocilia

201. Which tract can inhibit the vestibular reflex to the limb muscles?

- A. Medial vestibular tract
- B. Corticospinal tract
- C. Spinocerebellar tract
- D. Rubrospinal tract
- E. Reticulospinal tract

202. A test for presbycusis is done. A patient aged 45/60/90 starts hearing at which level?

- A. >90 dB
- B. 25-50 dB
- C. >50 dB
- D. <10 dB
- E. 10-25 dB

Mixed / Other Topics (49 questions)

203. A lumbar puncture consists of a needle being inserted through a space into the lumbar cistern, and aims to retrieve a sample of cerebrospinal fluid (CSF). Which of the following is the most likely level for the insertion of this needle?

- A. S2-S3
- B. L4-L5
- C. T12-L1
- D. T10-T11
- E. L1-L2

204. Space that connects between the lateral and the third ventricles is called . . .

- A. interventricular foramen
- B. foramen of Magendie
- C. cerebral aqueduct
- D. foramen of Luschka
- E. emissary foramen

205. Which of the following types of nerve fibers has the fastest conduction velocity?

- A. Group II fibers
- B. C (IV) fibers
- C. A- β (Ib) fibers
- D. A- δ (III) fibers
- E. A- α (Ia) fibers

206. The labyrinthine artery is an important source of blood supply to the inner ear. Which of the following arteries represents the major vessel from which this branch usually arises?

- A. Anterior inferior cerebellar
- B. Basilar
- C. Posterior inferior cerebellar
- D. Superior cerebellar
- E. Vertebral artery

207. Which of the following would be expected to see signs and symptoms of the patient from the lesion below?

- A. Nystagmus occurs normally in the right eye.
- B. Nystagmus occurs normally in the left eye.
- C. Horizontal conjugate eye movement occurs normally to the left.
- D. Horizontal conjugate eye movement occurs normally to the right.
- E. Abnormally conjugate eye movement to left with nystagmus to the right sides.

208. Which of the following sensation losses on the ipsilateral side of the lesion and at the only the level L2?

- A. Pain
- B. Pressure
- C. Crude touch
- D. Proprioception
- E. Fine touch

209. A patient was diagnosed with meningitis by lumbar puncture. Which of the following sites is the most commonly used for CSF collection?

- A. T10-T11
- B. T12-L1
- C. L1-L2
- D. L4-L5
- E. S2-S3

210. Which of the following receptor type receptors detect the tension of muscle and tendon? (only 4 choices)

- A. Nearby external receptor
- B. Telereceptors
- C. Proprioceptor
- D. Visceroceptor
- E. Chemoreceptor

211. Which of the following region is the most common of the encephalocele?

- A. Lumbar
- B. Frontal
- C. Occipital
- D. Temporal
- E. Parietal

212. A 43-year-old woman was sent to consult the doctor with drooping of right side of mouth, could not smile symmetrically. However, she can close both eye lids tightly and elevate her eyebrows () and forehead (). Where is the most likely specific site of lesion?

- A. Internal capsule
- B. Corticospinal tract
- C. Primary motor cortex
- D. Facial motor nucleus
- E. Cerebellum

213. A 65-year-old man has a pernicious anemia and memory problem. Which of the following compounds in blood is higher than normal and linked to memory problem in this man?

- A. folate
- B. creatinine
- C. methionine
- D. homocysteine
- E. Vitamin B12 level (elevated)

214. If the right Edinger Westphal nucleus is damaged from whatever, what is (are) the testing results?

- A. When light shines into Rt. eye, Rt pupil constricts, but Lt. not respond.
- B. When light shines into Rt. eye, Lt. pupil constricts, but Rt. not respond.
- C. When light shines into the left eye, the left and right pupils constrict.
- D. When light shines into left eye, both pupils do not constrict.
- E. No pupillary response occurs to any stimulus in either eye

215. Which fibers carry the first pain after injecting a needle into the skin? (only 4 choices)

- A. C fibers
- B. A δ (delta)
- C. A β (beta)
- D. A α (alpha)
- E. B fibers

216. Which of the following types of nerve fibers has the fastest conduction velocity?

- A. II
- B. C (IV)
- C. A- β (Ib)
- D. A- δ (III)
- E. A- α (Ia)

217. What type of transport is involved in the rabies virus, and by which specific molecular motor is it mediated?

- A. Anterograde - dynein mediated
- B. Anterograde - kinesin mediated
- C. Retrograde - dynein mediated
- D. Retrograde - kinesin mediated
- E. Retrograde - myosin mediated

218. Which of the following conditions is primarily characterised by the presence of plaques in the periventricular regions?

- A. Alzheimer's disease
- B. Parkinson's disease
- C. Hydrocephalus
- D. Multiple sclerosis
- E. Amyotrophic lateral sclerosis

219. What is the type of synapse between an axon and another axon called?

- A. Axo-dendritic
- B. Axo-somatic
- C. Axo-axonic
- D. Dendro-dendritic
- E. Somato-dendritic

220. What is the percent of the body's glucose used by the brain?

- A. 25%
- B. 50%
- C. 75%
- D. 100%
- E. 10%

221. What causes glutamate toxicity to happen?

- A. hypoglycemia
- B. hyperglycemia
- C. hyponatremia
- D. hypokalemia
- E. hyperthermia

222. What would be the result of a lesion to the Corticospinal tract in the pons?

- A. motor loss on the right side
- B. motor loss on the left side
- C. sensory loss on the left side
- D. sensory loss on the right side
- E. No deficit occurs

223. Which tract does not pass through the inferior cerebellar peduncle?

- A. ventral spinocerebellar
- B. olivocerebellar
- C. pontocerebellar
- D. dorsal spinocerebellar
- E. Cuneocerebellar tract

224. Mechanoreceptors of the skin transmit signals via which type of nerve fiber?

- A. A beta
- B. A alpha
- C. C fiber
- D. B fiber
- E. C fiber and A alpha combined

225. What doesn't synapse at the spinal trigeminal nucleus?

- A. pain temp from trigeminal
- B. pain temp from facial
- C. pain temp from glossopharyngeal
- D. pain temp from vagus
- E. pain temp from accessory

226. Which of the following is an epicritic sensation?

- A. pain
- B. temperature
- C. fine touch
- D. crude touch
- E. itch

227. What receptor is responsible for the sensory adaptation experienced when wearing a necklace or watch?

- A. Meissner's receptor
- B. Pacinian receptor
- C. Free nerve ending
- D. Krause receptor
- E. Peritrichial receptor (hair root plexus)

228. What transports glutamate into a glial cell?

- A. EAAT
- B. GLNT
- C. VGUT
- D. COMT
- E. MCT

- 229.** Transportation of Rabies occurred as which of the following directions and by which specific molecular motor?
- A. Anterograde - dynein mediated
 - B. Anterograde - kinesin mediated
 - C. Retrograde - dynein mediated
 - D. Retrograde - kinesin mediated
 - E. Anterograde - myosin mediated
- 230.** Which part of the brain specializes for motor control of our body, limb and eye movement?
- A. Limbic lobe
 - B. Thalamus
 - C. Frontal cortex
 - D. Hypothalamus
 - E. Occipital cortex
- 231.** Which of the following non-CNS structures is located in the Longitudinal fissure?
- A. Septum pellucidum
 - B. falx cerebri
 - C. tentorium cerebelli
 - D. falx cerebelli
 - E. Corpus callosum
- 232.** In normal condition, which of the following is used by a neuron to produce ATP?
- A. glucose
 - B. lactate
 - C. fumarate
 - D. glutamate
 - E. pyruvate
- 233.** What is the condition where meninges and brain tissue protrude?
- A. Cranium bifidum with meningoencephalocele
 - B. Cranium bifidum with meningocele
 - C. Cranium bifidum with meningoencephalocele
 - D. Cranium bifidum with encephalocele only
 - E. Spina bifida with myelomeningocele
- 234.** Which reflex test is appropriate for the C5-6 segment?
- A. Knee jerk
 - B. Triceps jerk
 - C. Biceps jerk
 - D. Jaw jerk
 - E. Brachioradialis jerk
- 235.** Neuroblasts from the alar plate can migrate ventrally to form which of the following structures?
- A. Pontine nucleus
 - B. Olivary nucleus
 - C. Motor nucleus of CN.7
 - D. Motor nucleus of 12
 - E. Motor nucleus of 6

236. What disease is characterized by the absence of iris? .

- A. Coloboma
- B. Aphakia
- C. Aniridia
- D. Cyclopia
- E. Microphthalmia

237. A 58-year-old patient has a drooping left eyelid and cannot wrinkle their forehead. What is the cause of the lesion?

- A. motor cn5
- B. motor cn7
- C. primary motor cortex
- D. sensory CN5
- E. CN 3 palsy

238. What structure is primarily stimulated when running forward (horizontal linear acceleration)?

- A. Utricle
- B. Saccule
- C. Superior semicircular canal
- D. Posterior semicircular canal
- E. Lateral semicircular canal

239. All statements are true about corneal epithelium 'except'

- A. Nourish by tears and aqueous humor
- B. maintain and ... by limbal cell
- C. High regeneration
- D. has a rich blood supply
- E. has many sensory receptors

240. Which nerve fiber is most sensitive to anesthesia?

- A. A alpha fiber
- B. A beta fiber
- C. C fiber
- D. B fiber
- E. A gamma fiber

241. A 25-year-old man came to the hospital. He dropped a pan on his legs (splashed with hot water). Which receptor is involved?

- A. TPRA1
- B. TPRV1
- C. TPRM8
- D. TPRV3
- E. TPRV4

242. A baby boy has a certain condition. The mother has no history of taking folic acid supplements and has no other abnormalities. Which disease is related?

- A. Holoprosencephaly
- B. Hydrocephalus
- C. Encephalocele
- D. Microcephaly
- E. Merocephaly

243. What receptor is related to ptosis and double vision?

- A. N1
- B. 5-T3
- C. D2
- D. NMDA
- E. AMPA

244. What is the target for Opioid drugs?

- A. Thalamus
- B. Hypothalamus
- C. Cerebellum
- D. Periaqueductal gray (PAG)
- E. Basal ganglia

245. A person turns their head to the right, but their eyes remain looking forward. The left eye moves to the left, while the right eye stays in the middle. When turning the head to the left, both eyes move to the right. Where is the abnormality?

- A. Right oculomotor
- B. Left abducens nucleus
- C. Left oculomotor
- D. Right abducens nucleus
- E. Left trochlear nerve

246. Which molecular motor is used for rabies transport?

- A. Dynein
- B. Kinesin
- C. Myosin
- D. Actin
- E. Tubulin

247. What is a gated-channel control?

- A. electrical synapse
- B. spatial summation
- C. temporal summation
- D. presynaptic facilitation
- E. presynaptic inhibition

248. Which of these is part of the paleocerebellum?

- A. vermis and anterior lobe
- B. flocculonodular lobe
- C. posterior lobe only
- D. anterior lobe only
- E. Dentate nucleus only

249. Which of the following is an affective/protopathic sensation?

- A. Vibration
- B. Pressure
- C. Pain
- D. Fine touch
- E. Two point discrimination

250. A patient with Benedikt syndrome would exhibit ipsilateral CN III palsy and what other sign?

- A. Contralateral loss of pain and temperature
- B. Contralateral hemiparesis
- C. Ipsilateral loss of fine touch and proprioception
- D. Ipsilateral cerebellar ataxia
- E. Bilateral lower limb weakness

251. A 40-year-old patient has a plaque in the periventricular area seen on MRI. What is the diagnosis?

- A. Alzheimer's disease
- B. Parkinson's disease
- C. Stroke (Cerebrovascular Accident)
- D. Multiple Sclerosis
- E. Epilepsy

Answer Key & Explanations

Lectures 1–16 (Exam 1 / FA-SA1) · 251 Questions · Single Best Answer

Lecture 1 — Organization of the CNS (3 questions)

Q1. Concerning the parts of Central Nervous System, which of the following items IS NOT included?

Correct answer: D. Trigeminal ganglion

The trigeminal ganglion sits in Meckel's cave and holds pseudounipolar sensory cell bodies — it's PNS tissue, not CNS. The pineal gland and posterior pituitary are both CNS despite sitting outside the main cranial vault, because they're direct neuroectodermal outgrowths of the diencephalon.

Q2. Concerning the parts of Central Nervous System, which of the following items IS NOT included?

Correct answer: D. Trigeminal ganglion

Same question as Q1 — trigeminal ganglion is PNS, not CNS.

Q3. When we wear a ring or watch, our body receives less stimulation from the central nervous system because of which receptor has...

Correct answer: B. Pacinian receptor

Pacinian corpuscles are rapidly-adapting (phasic) receptors — they fire at the onset of a stimulus and then stop, which is exactly why sustained pressure from a ring or watch fades from awareness.

Lecture 2 — Neurohistology (Neuron & Neuroglia) (19 questions)

Q4. Which of the following compounds is transferred from astrocyte to neuron and use for ATP production during starvation?

Correct answer: E. Lactate

This is the astrocyte-neuron LACTATE shuttle — astrocytes run glycolysis and export lactate (via MCT) to fuel neuronal oxidative metabolism, especially during glucose scarcity. The archive marked 'Citrate' (B), which is wrong; citrate is a TCA-cycle intermediate, not the shuttled fuel molecule.

♦ Corrected from the original archive extraction, which had marked B. Citrate — that was wrong.

Q5. Myelin sheath wrapping nerve axons in the basal ganglia is produced by which of the following cells?

Correct answer: D. Oligodendrocyte

Oligodendrocytes myelinate CNS axons (including basal ganglia); Schwann cells do the equivalent job only in the PNS.

Q6. Which of the following is the Glutamate transporter into astrocyte?

Correct answer: E. EAATs

Glutamate uptake INTO astrocytes is via EAATs (excitatory amino acid transporters, especially EAAT-2/GLT-1). The archive answer 'MCT' (monocarboxylate transporter) is wrong — MCT moves lactate/pyruvate/ketones, not glutamate.

♦ Corrected from the original archive extraction, which had marked A. MCT — that was wrong.

Q7. If the structure that carry fast pain is damaged, which of the following can be described as the structure that carry fast pain?

Correct answer: D. Neo-lateral spinothalamic tract

Fast, sharp, well-localized pain is carried by A-delta fibers via the neo(lateral)spinothalamic tract. The archive answer 'A-β fiber' is wrong — A-β fibers carry touch/pressure, not pain, and 13-20 μm diameter (option B) is actually the A-alpha range (motor/proprioception), not a pain fiber range either.

♦ Corrected from the original archive extraction, which had marked A. A-β fiber — that was wrong.

Q8. Which of the following compounds is transferred from astrocyte to neuron and use for ATP production during starvation?

Correct answer: B. lactate

Same underlying fact as Q4 — astrocytes shuttle lactate to neurons during starvation. The archive marked 'succinyl-CoA' (D), which is a TCA-cycle intermediate, not the shuttled molecule.

♦ Corrected from the original archive extraction, which had marked D. succinyl-CoA — that was wrong.

Q9. Which of the following items IS NOT a component of the blood-brain-barrier?

Correct answer: A. Oligodendrocyte

The neurovascular unit forming the BBB is built from endothelial tight junctions, the basement membrane, pericytes, and astrocyte end-feet. Oligodendrocytes play no structural role in it.

Q10. If the structure the carry fast pain damage, which of the following can be described the structure that carry fast pain?

Correct answer: D. Neo-lateral spinothalamic tract

Same core fact as Q7 (reworded, fewer options) — fast pain runs via the neo-lateral spinothalamic tract. This one was correctly extracted in the original archive.

Q11. Which of the following cells is associated with the immune system in the CNS?

Correct answer: A. Microglia cell

Microglia are the CNS's resident immune/phagocytic cells (mesoderm/yolk-sac derived), analogous to macrophages elsewhere in the body.

Q12. Which part of the axon has dense voltage-gated channels needed for Action potential?

Correct answer: A. Node of Ranvier

Voltage-gated Na⁺ channels cluster densely at the nodes of Ranvier, which is what enables fast saltatory conduction.

Q13. Which disease that immune system attacks and destroys myelin sheath?

Correct answer: B. Multiple sclerosis

Multiple sclerosis is the autoimmune disease that attacks and destroys CNS myelin. The archive answer 'Alzheimer disease' is clearly wrong — Alzheimer's pathology is amyloid plaques and neurofibrillary tangles, not autoimmune demyelination.

♦ Corrected from the original archive extraction, which had marked A. Alzheimer disease — that was wrong.

Q14. What is correct about sensory ganglia and autonomic ganglia?

Correct answer: A. Sensory ganglia are pseudounipolar, ANS ganglia are multipolar

Sensory ganglia (DRG, cranial sensory ganglia) contain pseudounipolar neurons; autonomic ganglia contain multipolar neurons that form a true synapse between pre- and postganglionic fibers.

Q15. Which is incorrect about the nervous system?

Correct answer: A. Neuron has glycogen synthase

This is the FALSE statement (matching what the question asks for). Neurons do NOT meaningfully store glycogen or run glycogen synthase — that job belongs to astrocytes, which is exactly why neurons depend on the astrocyte lactate shuttle in the first place.

Q16. Which of the following is the glutamate transporter to astrocytes?

Correct answer: E. EAATS

EAATs are the glutamate transporters into astrocytes.

Q17. Which of the following is the Lactate transporter to astrocytes?

Correct answer: B. MCT

MCT (monocarboxylate transporter) is the lactate transporter. The archive answer 'IAATS' (C) isn't a real, standard transporter name for this role — this looks like a garbled/incorrect extraction. MCT1/4 on astrocytes and MCT2 on neurons are the actual lactate shuttle transporters.

♦ Corrected from the original archive extraction, which had marked C. IAATS — that was wrong.

Q18. The structure around the nerve fiber (1st layer: from innermost) is

Archive answer: C. Neurolemma

FLAGGED — inconsistent with Q22, which asks nearly identical wording ('innermost structure around a single nerve fiber') and the archive answered 'Myelin sheath' there. Either both should be Myelin sheath, or the course specifically wants 'Neurolemma' (the Schwann cell's outer sheath, technically the true innermost investment for BOTH myelinated and unmyelinated fibers, since myelin only exists for myelinated ones). Verify with your instructor's exact framing — I'm not confident enough to override either direction.

■ Flagged as genuinely ambiguous, image/table-dependent, or having more than one defensible answer — verify against your source material.

Q19. Which of these is not part of the BBB (Blood-Brain Barrier)?

Correct answer: D. Ependymal cell

Ependymal cells line the ventricles/central canal — not part of the blood-brain barrier itself.

Q20. What is the 1st layer that covers an axon (from inside) in the PNS?

Correct answer: B. Myelin sheath

The myelin sheath is the first/innermost layer directly covering a myelinated PNS axon.

Q21. Which statement is correct about a neuron?

Correct answer: D. neuron has abundant of rER

Neurons have abundant rough ER, visible histologically as Nissl bodies/substance — a hallmark of their high protein-synthesis demand.

Q22. The innermost structure around a single nerve fiber?

Correct answer: B. Myelin sheath

Same core question as Q18 but this one was extracted consistently as 'Myelin sheath' — see the Q18 flag above for the inconsistency between these two near-identical questions.

Lecture 3 — Neuron Properties: RMP & Action Potential (10 questions)

Q23. A 40-year-old male presented with hand injury needs immediate suture. Lidocaine is given as a local anesthesia. What is the...

Correct answer: C. Inhibition of voltage-gated sodium channel

Lidocaine (an amide local anesthetic) blocks voltage-gated Na⁺ channels, preventing action potential propagation in nociceptive fibers.

Q24. Hyperpolarization of the postsynaptic membrane in response to glycine or GABA is produced by the opening of. . .

Correct answer: B. Cl⁻ channels

GABA-A and glycine receptors are both ionotropic Cl⁻ channels; Cl⁻ influx hyperpolarizes the postsynaptic membrane.

Q25. A 35-year old male was given a lidocaine injection prior to incision and drainage of the abscesses () at right buttocks. Which...

Correct answer: C. Block voltage-gated Na⁺ channels

Same lidocaine mechanism as Q23 — voltage-gated Na⁺ channel blockade.

Q26. A 40-year-old male presented with hand injury need immediate suture. Lidocaine is given as a local anesthesia. What is the...

Correct answer: D. Inhibition of voltage-gated sodium channel

Same lidocaine mechanism, different option lettering.

Q27. Muscle weakness in the afternoon and evening. No symptoms in the morning. What is the pathogenesis of this disease [NL 2024]?

Correct answer: A. Autoantibody to Ach receptor

Fatigable weakness that worsens through the day (and improves with rest overnight) is classic myasthenia gravis — autoantibodies against postsynaptic nicotinic ACh receptors.

Q28. What is the major energy usage in the brain?

Archive answer: B. resting potential

FLAGGED — genuinely debated framing. The classic teaching (Attwell & Laughlin's brain energy budget analysis) is that the single largest share of brain ATP goes to restoring ionic gradients dissipated by ACTION POTENTIALS and synaptic signaling, which would favor 'Action potential' (A) over 'resting potential' (B, the archive's answer). But maintaining the resting potential itself also runs continuously via the Na⁺/K⁺-ATPase even without firing, so depending on exactly how your lecture framed this, B could also be defensible. Verify against your specific slide.

■ Flagged as genuinely ambiguous, image/table-dependent, or having more than one defensible answer — verify against your source material.

Q29. Use Na⁺/K⁺ pumping during?

Archive answer: D. resting stage

FLAGGED — related ambiguity to Q28. If this is testing 'the Na⁺/K⁺ pump runs continuously, even at rest, not just during spikes,' then D (resting stage) is a reasonable intended answer. If it's testing 'the pump's main job is restoring gradients after firing,' the expected answer would look more like repolarization/hyperpolarization. Verify against your specific slide's framing.

■ Flagged as genuinely ambiguous, image/table-dependent, or having more than one defensible answer — verify against your source material.

Q30. What is it called when depolarization from multiple synapses summates?

Correct answer: C. spatial summation

Spatial summation = inputs from DIFFERENT synapses/locations arriving together add up. (Temporal summation would be repeated inputs at the SAME synapse in rapid succession — not what's described here.)

Q31. All of the following are correct about chemical synapse EXCEPT which one?

Correct answer: C. No synaptic delay

This is the FALSE statement. Chemical synapses always have a measurable synaptic delay (~0.5–1 ms) from vesicle fusion and diffusion — 'no synaptic delay' describes electrical synapses (gap junctions), not chemical ones.

Q32. Which statement about graded potentials is incorrect?

Correct answer: D. They transmit signals over long distances without decreasing in strength.

This is the FALSE statement. Graded potentials are explicitly decremental — they lose strength with distance. Signals that travel long distances WITHOUT decrementing are action potentials, not graded potentials.

Lecture 4 — Metabolic Pathways of Neurons (5 questions)

Q33. Which of the followings is INCORRECT?

Correct answer: D. Astrocyte can convert reduced ascorbic acid to oxidized ascorbic acid

This is the FALSE statement (matching 'INCORRECT'). Astrocytes recycle ascorbate in the OPPOSITE direction — oxidized ascorbic acid back to reduced ascorbic acid (using NADPH from the pentose phosphate pathway) — as part of antioxidant defense. The statement as written reverses that direction, making it false.

Q34. Which of the followings is INCORRECT?

Correct answer: D. Glycogen synthase is the most active enzyme and found in neuron.

This is the FALSE statement. Neurons do NOT have high glycogen synthase activity or meaningful glycogen stores — that's an astrocyte function.

Q35. What enzyme hydrolyzes glutamine to glutamate? (MD27,28)

Correct answer: A. Glutaminase

Glutaminase converts glutamine back to glutamate inside neurons, completing the glutamate-glutamine cycle.

Q36. Accumulation of glutamate causes toxicity due to the accumulation of what ion in the cell?

Correct answer: A. Calcium

Excess extracellular glutamate over-activates NMDA/AMPA receptors, driving toxic Ca²⁺ influx into the postsynaptic neuron — the core mechanism of excitotoxicity.

Q37. What enzyme hydrolyzes glutamine to glutamate?

Correct answer: A. Glutaminase

Same as Q35 — glutaminase.

Lecture 5 — Embryonic Development of the NS (12 questions)

Q38. Which of the following structures developed to be substantia nigra?

Correct answer: E. Mesencephalon

Substantia nigra is a midbrain structure, i.e. mesencephalon-derived (E). The archive answer 'Myelencephalon' (B) is wrong — myelencephalon becomes the medulla, a completely different brainstem level.

♦ Corrected from the original archive extraction, which had marked B. Myelencephalon — that was wrong.

Q39. The optic cup is an evagination of which of the following?

Correct answer: B. Diencephalon

The optic vesicle (which invaginates to form the optic cup) is an outgrowth of the diencephalon (B), not the mesencephalon (C). This is a clear error in the archive.

♦ Corrected from the original archive extraction, which had marked C. Mesencephalon — that was wrong.

Q40. Which of the following diseases have the characteristic “protrusion of meninges”?

Correct answer: D. Spina bifida with meningocele

Meningocele = meninges (and CSF) protrude through a vertebral defect WITHOUT neural tissue — this is exactly 'protrusion of meninges.' Meningomyelocele would add spinal cord tissue too, which isn't what the question describes.

Q41. Which of the following structures developed to the pineal gland?

Correct answer: A. Diencephalon

The pineal gland develops from the roof of the diencephalon (epithalamus) (A), not the telencephalon (B). Clear archive error.

♦ Corrected from the original archive extraction, which had marked B. Telencephalon — that was wrong.

Q42. Which of the following represents the general somatic efferent (GSE) at the myelencephalon?

Correct answer: C. Hypoglossal nucleus

The hypoglossal nerve (CN XII, nucleus in the myelencephalon/medulla) is a pure motor nerve to tongue muscles — classified GSE, like the somatic-motor nuclei of III, IV, VI.

Q43. Which of the following structures differentiates into the optic cup?

Correct answer: C. Neuroectoderm

The optic cup is neuroectoderm (C), an outpouching of the diencephalon — not mesoderm (B). Clear archive error.

♦ Corrected from the original archive extraction, which had marked B. Mesoderm — that was wrong.

Q44. What birth defect of the spinal cord is associated with hydrocephalus?

Correct answer: A. meningocele

Myelomeningocele is strongly linked to Arnold-Chiari II malformation, which obstructs CSF flow and causes hydrocephalus.

Q45. Which condition results from the failure of neural folds to fuse?

Correct answer: C. myeloschisis

Myeloschisis describes a completely open, unfused neural plate — the most direct literal match for 'failure of the neural folds to fuse.'

Q46. What is the developmental origin of the pituitary gland?

Correct answer: A. Diencephalon

Among the three options given, Diencephalon is correct for the posterior pituitary (neurohypophysis). Note the full picture: the pituitary actually has a DUAL origin — Rathke's pouch (surface ectoderm) forms the anterior lobe, and a diencephalic downgrowth forms the posterior lobe. If 'surface ectoderm' had been an option, that nuance would matter more.

Q47. What does corneal endothelium develop from?

Correct answer: C. neural crest cells

Corneal endothelium is a neural crest derivative (migrating periocular mesenchyme) — distinct from the corneal EPITHELIUM, which is surface ectoderm.

Q48. What is the condition from failure in Anterior neuropore closure?

Correct answer: B. Meroencephaly

Meroencephaly is another name for anencephaly, which results from failure of the anterior neuropore to close.

Q49. What does the lens develop from?

Correct answer: A. Surface ectoderm

The lens develops from surface ectoderm via the lens placode, induced by the underlying optic vesicle.

Lecture 6 — Development of the Eye (4 questions)

Q50. High intracerebral pressure resulting in which of the following signs and symptoms?

Correct answer: A. Papillary edema

Papilledema (optic disc swelling from raised ICP transmitted along the optic nerve sheath) is the classic sign of high intracranial pressure.

Q51. What is derived from the ventral part of the optic vesicle?

Archive answer: A. Choroid fissure

FLAGGED — 'choroid fissure' and 'retinal fissure' are, in most standard embryology texts, synonyms for the same groove on the ventral optic stalk/cup (through which the hyaloid vessels pass). Presenting them as two different answer options is unusual; whichever term your specific course slide uses as 'the' term is the one to go with — I can't meaningfully distinguish them as separate correct/incorrect answers.

■ Flagged as genuinely ambiguous, image/table-dependent, or having more than one defensible answer — verify against your source material.

Q52. Failure of the choroid fissure to fuse results in?

Correct answer: A. Coloboma

Failure of the choroid/retinal fissure to fuse causes a coloboma (A) — a persistent gap/cleft in the iris, retina, or optic disc. The archive answer 'Anophthalmia' (B) is wrong; anophthalmia is complete absence of the eye from failure of optic vesicle formation in the first place, an earlier and different problem.

♦ Corrected from the original archive extraction, which had marked B. Anophthalmia — that was wrong.

Q53. The spinal trigeminal nucleus is derived from what?

Correct answer: B. Alar part

Sensory brainstem nuclei (including the spinal trigeminal nucleus) derive from the alar plate (dorsal, sensory-associated); motor nuclei derive from the ventral basal plate.

Lecture 7 — Spinal Cord & Blood Supply (16 questions)

Q54. The result of the MRI exhibits hemisection of spinal cord (Brown sequard syndrome) at the right L2 spinal nerve level. Which...

Correct answer: C. Conscious proprioception

The question asks what's lost on the SAME (ipsilateral) side at and below the lesion. In Brown-Séquard, ipsilateral losses are dorsal-column functions (proprioception, vibration) and motor — because those tracts haven't crossed yet at that level. Pain and temperature are LOST CONTRALATERALLY (the spinothalamic tract has already crossed), so 'Pain' (the archive's answer) is actually describing the wrong side. The correct ipsilateral loss is C, Conscious proprioception.

♦ Corrected from the original archive extraction, which had marked B. Pain — that was wrong.

Q55. A 65-year-old man has lost vibration, position and joint discrimination in the legs and lower body. Lesions to which of the...

Correct answer: B. Dorsal column (funiculus)

Vibration, position, and joint discrimination are dorsal column (funiculus) functions.

Q56. Stiff neck is a compensated sign to limit head and neck movements of a patient who is likely affected at which of the...

Correct answer: D. Meningeal dura

Nuchal rigidity is a protective reflex against stretching inflamed/irritated meninges (dura).

Q57. A patient has an injury that damages the lateral horn of the spinal cord. Which of the following would you expect to see in...

Correct answer: C. Loss of sympathetic impulse

The lateral horn (thoracolumbar levels) houses preganglionic sympathetic neurons — damage disrupts sympathetic outflow from that segment.

Q58. All of the following statements concerning the anterior spinal artery are correct EXCEPT

Correct answer: E. Occlusion would cause loss of proprioception

This is the FALSE statement. Anterior spinal artery occlusion SPARES proprioception (supplied instead by the paired posterior spinal arteries) — saying occlusion 'would cause loss of proprioception' is exactly backwards, which is why it's correctly flagged as the EXCEPT/false answer.

Q59. Stiff neck is a compensated sign to limit head and neck movements of a patient who is likely irritated at which of the...

Correct answer: D. Meningeal dura

Same as Q56.

Q60. A 53-year-old male was hospitalized for an acute spinal cord infarction. He presented with bilateral motor dysfunction, loss...

Correct answer: C. Anterior spinal cord syndrome

Bilateral motor loss + bilateral pain/temperature loss + PRESERVED proprioception/vibration is the textbook definition of anterior spinal cord syndrome (C) — the anterior spinal artery supplies the anterior 2/3 of the cord (motor + spinothalamic), sparing the dorsal columns (posterior spinal artery territory). The archive answer 'Central cord syndrome' (D) is wrong; central cord syndrome instead classically shows arm-greater-than-leg weakness with variable, patchy sensory loss, not this clean anterior-syndrome pattern.

♦ Corrected from the original archive extraction, which had marked D. Central cord syndrome — that was wrong.

Q61. The result of MRI exhibits hemisection of the spinal cord (Brown Sequard syndrome) at the Right L2 spinal cord level. Which...

Correct answer: D. Contralateral loss of pain and temperature from the L4 and below

Contralateral spinothalamic (pain/temperature) loss below the lesion level is the classic Brown-Séquard pattern.

Q62. A 55-year-old man suffered an injury that caused damage to the dorsal column of his spinal cord. Which of the following signs...

Correct answer: B. Loss of proprioceptive

Dorsal column damage causes loss of proprioception (and vibration/fine touch).

Q63. Stiff neck is a compensated sign to limit head and neck movements of a patient who is likely irritated at which of the...

Correct answer: D. Meningeal dura

Same as Q56/59.

Q64. What function is lost with damage to the lateral horn of the spinal cord?

Correct answer: A. Intermediolateral nucleus

The lateral horn (as generally taught, thoracolumbar T1-L2/3) is the intermediolateral nucleus (A) — the column of preganglionic sympathetic neurons. 'Sacral parasympathetic nucleus' (B, the archive's answer) is a related but distinct structure found specifically at S2-S4, not what 'the lateral horn' refers to without further qualification.

♦ Corrected from the original archive extraction, which had marked B. Sacral parasympathetic nucleus — that was wrong.

Q65. What is a likely consequence of posterior spinal cord artery damage?

Correct answer: C. loss of vibration and proprioception

Posterior spinal artery territory is the dorsal columns — damage causes loss of vibration and proprioception (C), not leg weakness (A, the archive's answer). Motor weakness would instead be an ANTERIOR spinal artery finding (anterior 2/3 of the cord, including the corticospinal tracts).

♦ Corrected from the original archive extraction, which had marked A. leg weakness — that was wrong.

Q66. Which of the following is not part of the CNS?

Correct answer: D. vagal ganglion

A ganglion (vagal ganglion) sits outside the CNS by definition — it's a PNS structure.

Q67. If the anterior spinal artery is damaged, what is spared?

Archive answer: C. crude touch 37

FLAGGED — both C (crude touch) and E (proprioception) are defensible 'spared' answers for anterior spinal artery occlusion. Proprioception/vibration sparing is the unambiguous, universally-taught fact (dorsal columns have separate blood supply). Crude/light touch sparing is also sometimes taught, based on touch having redundant pathways (partially carried by the intact dorsal columns even when the anterior spinothalamic component is knocked out). If your course emphasizes the dorsal-column-sparing fact primarily, the 'intended' answer may actually be proprioception rather than crude touch — worth double-checking your slide.

■ Flagged as genuinely ambiguous, image/table-dependent, or having more than one defensible answer — verify against your source material.

Q68. Which of the following is a primary motor symptom associated with a complete spinal cord lesion?

Correct answer: C. Tetraplegia or Paraplegia

A COMPLETE spinal cord lesion causes bilateral loss below the level — tetraplegia (if cervical) or paraplegia (if thoracic/lumbar/sacral) (C). The archive answer 'Unilateral reflex absence' (E) doesn't fit a complete (bilateral) lesion at all — that finding would instead suggest a partial/hemisecton lesion.

♦ Corrected from the original archive extraction, which had marked E. Unilateral reflex absence — that was wrong.

Q69. What extends from the pia mater to the dura mater, and separates the dorsal and ventral nerve roots?

Correct answer: D. denticulate ligament

The denticulate ligaments are lateral, tooth-like pial extensions piercing the arachnoid to attach to the dura, anatomically separating the dorsal and ventral nerve roots.

Lecture 8 — PNS: Cranial & Spinal Nerves (16 questions)

Q70. Which of the functional components below is found in the hypoglossal nucleus?

Correct answer: E. GSE

CN XII (hypoglossal) is pure motor to tongue muscles — classified GSE.

Q71. According to the Peripheral Nervous System (PNS), which of the statements below is INCORRECT?

Archive answer: E. There are 12 pairs of cranial nerves, and only CN. III - XII emerge from the brainstem.

FLAGGED — as written, statement E ('12 pairs of cranial nerves, only CN III-XII emerge from the brainstem') actually reads as TRUE to me — CN I and II don't emerge from the brainstem in the conventional sense, so III-XII (10 nerves) is accurate. If E is true, it can't be the 'INCORRECT' answer the question is looking for. The more likely candidate for the truly false statement is C ('only 1 functional component in the accessory nerve') IF your course teaches CN XI as having a cranial root (overlapping with vagus) plus a distinct spinal root — some frameworks would call that 2 components, not 1. Worth checking which framing your slides use.

■ Flagged as genuinely ambiguous, image/table-dependent, or having more than one defensible answer — verify against your source material.

Q72. Collections of nerve cell bodies inside the peripheral nervous system are called:

Correct answer: C. ganglia

Ganglia = PNS cell body collections (the CNS equivalent term is 'nucleus').

Q73. Which of the functional components IS NOT found in the facial nerve?

Correct answer: C. GSE

Facial muscles are branchiomeric (2nd pharyngeal arch), so CN VII's motor component is SVE, not GSE — GSE is the one NOT found in the facial nerve.

Q74. A 41-year-old woman complains to her family physician about recurring episodes of sharp pain that seem to originate from...

Correct answer: D. Trigeminal (V)

Sharp, severe, trigger-able facial/perioral pain is classic trigeminal neuralgia.

Q75. Which of the following spinal nerves are NOT included in a nerve plexus?

Correct answer: C. T2-T12

Most thoracic nerves (roughly T2-T12) run individually as intercostal nerves rather than forming a plexus.

Q76. When the light gets into the eye, which structures absorb the light?

Correct answer: A. Choroid

The heavily pigmented choroid absorbs stray light, reducing internal glare/reflection — like the black interior of a camera.

Q77. Which of the following spinal nerves are NOT included in a nerve plexus?

Correct answer: C. T2-T12

Same as Q75.

Q78. Which of the functional component of nerve fibers IS NOT found in a spinal nerve?

Correct answer: E. SVE

SVE (special visceral efferent, branchiomeric muscle motor) is exclusive to certain cranial nerves — spinal nerves never carry it.

Q79. Collections of nerve cell bodies (soma) inside the peripheral nervous system are called

Correct answer: D. Ganglia

Same definition as Q72.

Q80. Which of the functional components IS NOT found in the facial nerve?

Correct answer: C. GSE

Same as Q73.

Q81. Which nucleus contains General Visceral Efferent (GVE) fibers?

Correct answer: B. Dorsal motor nucleus of Vagus

The dorsal motor nucleus of the vagus is a genuine GVE (parasympathetic) nucleus, supplying thoracic/abdominal viscera. Note the inferior salivatory nucleus (option D) is ALSO a true GVE nucleus (CN IX, parotid gland) — both are legitimately correct facts, so if your source lists a different 'best' answer, it may be testing a finer distinction than what's visible from the question stem alone.

Q82. Which of the following is correct about the transverse section of caudal pons?

Archive answer: C. MLF is in the tectum above the fourth ventricle

FLAGGED, moderate confidence the archive is wrong. 'MLF is in the tectum above the fourth ventricle' (C, the archive's answer) is anatomically off — the MLF sits in the pontine TEGMENTUM near the floor of the 4th ventricle, and 'tectum' isn't standard terminology for the pons (tectum specifically refers to the midbrain roof). Statement A ('pontine nuclei lie between/among the transverse pontine fibers') is a more textbook-accurate description of ventral pons anatomy. Worth double-checking against your specific cross-section diagram.

■ Flagged as genuinely ambiguous, image/table-dependent, or having more than one defensible answer — verify against your source material.

Q83. An optical disc cannot detect light energy because of the absence of?

Correct answer: D. Absent Photoreceptor

The optic disc has no photoreceptors at all — it's purely the exit point for the optic nerve and central retinal vessels, which is why it's the physiologic blind spot.

Q84. What dermatome is listed incorrectly?

Correct answer: B. Posterior of thigh and leg – T1-2

This is the INCORRECTLY listed dermatome (matching what the question asks for). Posterior thigh/leg is not T1-2 — that level corresponds more to the medial arm/axilla region; posterior thigh/leg is much further down the dermatome ladder (roughly S1-S2/L-S region).

Q85. Which nerve fiber type is responsible for increased salivation & constriction of the urinary bladder?

Correct answer: E. General Visceral Efferent (GVE)

Both increased salivation (parasympathetic) and bladder sphincter effects are autonomic — GVE.

Lecture 9 — Brainstem Structures (17 questions)

Q86. The MRI result appears compression of a tumor at the right tuberculum cinerium of close medulla. Which of the following...

Correct answer: C. Pain and temperature of the right face

The tuberculum cinereum overlies the spinal trigeminal nucleus/tract — compression here disturbs ipsilateral facial pain/temperature.

Q87. The MRI result appears compression of a tumor at Right Tuberculum cinerium of close medulla. Which of the following structures...

Correct answer: B. Spinal trigeminal nucleus and tract

Tuberculum cinereum is the surface landmark overlying the spinal trigeminal nucleus and tract (B). The archive answer 'Gracile nucleus and fasciculus gracilis' (C) is wrong — that's a different medullary landmark entirely (the clava, further medial/dorsal).

♦ Corrected from the original archive extraction, which had marked C. Gracile nucleus and fasciculus gracilis — that was wrong.

Q88. Which of the following reflex tests represents functions of the medulla oblongata?

Correct answer: A. Gag reflex

The gag reflex (CN IX afferent, CN X efferent) tests medulla oblongata function.

Q89. A 20-year-old man w/ loss of muscle power at the left side of the body and his right eye has strabismus (). Pupillary light...

Correct answer: C. Rostral midbrain, right side

Ipsilateral CN III palsy (mydriasis, ptosis, strabismus) + contralateral body weakness = crossed Weber syndrome, localizing to the rostral midbrain on the side of the eye finding.

Q90. The MRI result appears compression of a tumor at Right tuberculum cinereum of close medulla. Which of the following nerves...

Correct answer: E. Accessory nerve

The spinal trigeminal nucleus (tuberculum cinereum) receives GSA fibers from CN V (which forms it directly), plus smaller GSA contributions from CN VII, IX, and X. CN XI (Accessory) is purely motor and contributes no sensory fibers here — so Accessory (E) is the one that does NOT synapse there. The archive answer 'Trigeminal nerve' (A) is clearly wrong, since trigeminal fibers are literally what forms this nucleus.

♦ Corrected from the original archive extraction, which had marked A. Trigeminal nerve — that was wrong.

Q91. Which of the reflex tests represents functions of the medulla oblongata?

Correct answer: A. Gag reflex

Same as Q88.

Q92. Which of the following neuronal pathways performs voluntary control of body and limb movements?

Correct answer: E. Pyramidal tract

The pyramidal (corticospinal) tract is THE pathway for voluntary control of body/limb movement. The archive answer 'Hypothalamospinal tract' (D) is wrong — that's a descending autonomic pathway, unrelated to voluntary movement.

♦ Corrected from the original archive extraction, which had marked D. Hypothalamospinal tract — that was wrong.

Q93. Which of the following CNS structures locates at the same horizontal level of tentorial cere- belli?

Correct answer: B. Midbrain

The midbrain passes through the tentorial incisura (opening in the tentorium cerebelli) — the anatomical basis of uncal/transtentorial herniation compressing the midbrain first.

Q94. A 32-year-old man loss of muscle power, presented Babinski sign at the left side. His right eye has lateral strabismus () and...

Correct answer: C. Rostral midbrain, right side

This describes the same crossed pattern as Q89 (CN III-type findings — lateral strabismus from unopposed lateral rectus, plus mydriasis — combined with contralateral body weakness/Babinski) = Weber syndrome = rostral midbrain, same side as the eye findings. The archive answer 'Mid pons, right side' (A) is inconsistent with the near-identical Q89, which correctly answered 'Rostral midbrain, right side.'

♦ Corrected from the original archive extraction, which had marked A. Mid pons, right side — that was wrong.

Q95. MRI result appears compression of a tumor at Right tuberculum cinereum of close medulla. Which of the following nerves DOES...

Correct answer: C. Accessory

Same underlying fact as Q90 (just re-ordered options) — Accessory nerve (C in this ordering) is the one that does NOT contribute fibers to the spinal trigeminal nucleus. The archive answer 'Vagus' (A) is wrong; vagus does contribute a small GSA component there.

♦ Corrected from the original archive extraction, which had marked A. Vagus — that was wrong.

Q96. Which of the reflex tests represents functions of the medulla oblongata?

Correct answer: A. Gag reflex

Same as Q88/91.

Q97. A patient presents with left eye symptoms where the pupil is larger than the right, and a history of lower facial weakness....

Archive answer: A. Midbrain

FLAGGED — the vignette combines a CN III-type pupil finding with 'lower facial weakness' (more of a CN VII/pontine or corticobulbar finding), which doesn't cleanly localize to one single classic brainstem level among the options given (Midbrain/Pons/Medulla/Cortex). 'Midbrain' explains the pupil finding but not clearly the facial weakness. Worth checking the original source material for the intended reasoning here.

■ Flagged as genuinely ambiguous, image/table-dependent, or having more than one defensible answer — verify against your source material.

Q98. A patient is diagnosed with loss of pain and temperature contralateral from the body and ipsilateral from the face. Where is...

Correct answer: B. Lateral medulla oblongata

Contralateral body pain/temperature loss + ipsilateral face pain/temperature loss is the classic lateral medullary (Wallenberg) syndrome pattern — localizing to the lateral medulla oblongata (B), not caudal pons (C, the archive's answer).

♦ Corrected from the original archive extraction, which had marked C. Caudal pons — that was wrong.

Q99. Which tract's secondary neuron synapses and crosses to the opposite side at the medulla oblongata?

Correct answer: C. Lateral corticospinal tract

This question was illegible/unmarked in the original archive. The lateral corticospinal tract (C) is the one whose fibers cross at the medulla (pyramidal decussation, ~90% of fibers). The lateral spinothalamic tract instead crosses much earlier, within 1-2 segments of entering the spinal cord — not at the medulla.

♦ Corrected from the original archive extraction, which had marked None. (unmarked/illegible) — that was wrong.

Q100. Damage to which major component of the Ascending Reticular Activating System (ARAS) would most directly affect consciousness?

Correct answer: D. Pons and Midbrain

The ARAS core is the pontomesencephalic reticular formation — bilateral damage here (or to its thalamic/cortical targets) is a classic cause of coma.

Q101. What is the major site of the ARAS related to coma?

Correct answer: C. Pons and midbrain

Same core fact as Q100.

Q102. A male patient presents with alteration of consciousness, irregular breathing, and abnormal limb posturing. Lesion in which of...

Archive answer: A. Cerebrum

FLAGGED, moderate confidence the archive is wrong. Altered consciousness + irregular breathing + abnormal limb posturing together is the classic triad for BRAINSTEM (reticular formation) dysfunction — most directly implicating the pons/midbrain, where ARAS, respiratory pattern centers, and descending motor pathways for posturing all coexist. The archive answer 'Cerebrum' (A) isn't wrong in every interpretation (a large cerebral lesion CAN secondarily cause this triad via herniation/brainstem compression), but if the question is asking for the most DIRECT localization, Midbrain (B) is the classically-taught answer.

■ *Flagged as genuinely ambiguous, image/table-dependent, or having more than one defensible answer — verify against your source material.*

Lecture 10 — Reticular Formation (3 questions)

Q103. Which of the following pairs is the correct match (related functions)?

Correct answer: B. Lateral spinothalamic tract – substantia gelatinosa

The substantia gelatinosa (Rexed lamina II) is where pain afferents first synapse before crossing to form the lateral spinothalamic tract — a genuine anatomical pairing. The other three options mismatch tract and nucleus.

Q104. Which of the following structures makes the fiber synapse with the motor nuclei of CN V and then to the tensor tympani muscle?

Correct answer: E. Reticular formation

The medial (central) reticular formation column provides the local interneuronal connections linking sensory input to cranial motor nuclei for reflexes like the acoustic (stapedial/tensor tympani) reflex.

Q105. Where is the ARAS not sent to? ■■■■■■■■■■

Correct answer: E. Cerebellar deep nuclei

The original answer options were illegible/redacted in the source archive, so they've been reconstructed based on standard ARAS anatomy. The ARAS projects to the thalamus (intralaminar nuclei), basal forebrain, hypothalamus, and diffusely to the cerebral cortex via thalamic relay — the cerebellar deep nuclei are not one of its primary/defining target pathways.

❖ *Corrected from the original archive extraction, which had marked None. (unmarked/illegible) — that was wrong.*

Lecture 12 — Somatosensory Pathways (8 questions)

Q106. Which of the following is correct about the Brown Sequard syndrome at Rt. Hemisection of L2?

Correct answer: E. Contralateral loss of pain and temperature from the L4 and below

Same Brown-Séquard pattern as Q61 — contralateral pain/temperature loss below the lesion.

Q107. Which neuronal tract/fasciculus passes the anterior white commissure?

Correct answer: D. Lateral spinothalamic tract

The lateral spinothalamic tract's second-order fibers cross via the anterior white commissure, within 1-2 segments of entering the cord.

Q108. Loss of proprioception on the left and loss of pain/temp on the right is likely to be?

Correct answer: A. brown sequard syndrome

Ipsilateral proprioception loss + contralateral pain/temperature loss is THE textbook definition of Brown-Séquard syndrome (A) — this exact dissociated pattern. The archive answer 'central cord syndrome' (B) is wrong; central cord syndrome instead shows a different pattern (bilateral, cape-like, arm-greater-than-leg).

◆ Corrected from the original archive extraction, which had marked *B. central cord syndrome* — that was wrong.

Q109. Which tract lesion would give a positive Romberg test result?

Correct answer: A. Dorsal column

A positive Romberg (steady eyes-open, unsteady eyes-closed) indicates a sensory (dorsal column/proprioceptive) deficit that vision was compensating for.

Q110. Which one affects the PNS?

Correct answer: C. Carpal tunnel syndrome

Carpal tunnel syndrome is a peripheral nerve (median nerve) compression — a PNS process. Alzheimer's and Brown-Séquard are both CNS.

Q111. A patient has right-sided pain and temperature loss from T12, left-sided pain and temperature loss from T10, left-sided...

Correct answer: B. brown sequard at T10

This complex vignette fits a LEFT-sided Brown-Séquard hemisection at T10: the ipsilateral (left) T10-level pain/temp loss reflects a narrow segmental band right at the lesion (before those specific entering fibers have crossed), the contralateral (right) pain/temp loss below (~T12) reflects the classic below-level spinothalamic pattern, and the ipsilateral proprioception loss plus ipsilateral Horner's both fit a left-sided lesion disrupting the dorsal columns and the descending sympathetic pathway together.

Q112. What structure for unconscious proprioception at S2-4 synapses at Clarke's nucleus?

Archive answer: A. Dorsal spinocerebellar

FLAGGED — Clarke's nucleus proper only extends roughly C8-L2/3, so it technically isn't present at S2-4 to synapse onto directly; unconscious proprioception from the sacral cord instead relays via an alternate route (functionally similar 'spinal border cell' input feeding the ventral spinocerebellar tract). If your source simplifies this and just wants 'dorsal spinocerebellar tract' as the general answer, A stands; if it's specifically testing the S2-4 exception, the intended answer may differ. Worth checking your slide's exact wording.

■ Flagged as genuinely ambiguous, image/table-dependent, or having more than one defensible answer — verify against your source material.

Q113. Which unconscious tract synapses with Clarke's nucleus?

Correct answer: D. dorsal spinocerebellar

The dorsal spinocerebellar tract's second-order neurons ARE the Clarke's nucleus neurons (for the levels where Clarke's nucleus exists, C8-L2/3).

Lecture 13 — Taste (Gustatory) Sensation (6 questions)

Q114. According to the lesion in a providing picture, what are the expected signs and symptoms of this lesion?

Archive answer: E. Tongue deviates to the left side, Right hemiplegia, Loss of light touch and proprioception senses at right side

FLAGGED — this question references an image ('providing picture') that isn't included in the extracted text, so I can't independently verify against the actual diagram. The stated answer (tongue left, right hemiplegia, right-sided touch/proprioception loss) IS internally consistent with a classic medial medullary (Dejerine) syndrome — ipsilateral CN XII palsy (tongue deviates toward the lesion) plus contralateral corticospinal and medial lemniscus involvement — so it's plausible, just unverifiable without the picture.

■ Flagged as genuinely ambiguous, image/table-dependent, or having more than one defensible answer — verify against your source material.

Q115. Autonomic controlling of the heart, lungs, digestion, taste, communication between brain and organs is..... nerves

Correct answer: B. Vagus

The vagus nerve carries the widest autonomic reach of any cranial nerve — heart, lungs, GI tract, plus taste from the epiglottis.

Q116. A 71-year-old man complains to his family physician that his face 'feels funny.' The examination reveals numbness on his...

Correct answer: A. Lateral one-third of the postcentral

Face and tongue sensory representation sits in the lateral one-third of the postcentral gyrus (per the sensory homunculus).

Q117. According to the lesion in a providing picture, what are the expected signs and symptoms of this lesion?

Archive answer: A. Tongue deviates to the left side, Right hemiplegia, Loss of light touch and proprioception senses from right side.

FLAGGED — same image-dependent situation as Q114; internally consistent with medial medullary syndrome logic but unverifiable without the referenced picture.

■ Flagged as genuinely ambiguous, image/table-dependent, or having more than one defensible answer — verify against your source material.

Q118. Autonomic controlling of the heart, lungs, digestion and taste communication between brain and organs is..... nerve.

Correct answer: A. Vagus

Same as Q115.

Q119. What is the correct pathway for taste sensation from the anterior 2/3 of the tongue?

Correct answer: A. Facial nerve (Chorda tympani) → Geniculate ganglion → Solitary nucleus (upper part) → VPM of thalamus → Gustatory area of cerebral cortex.

CN VII (chorda tympani) → geniculate ganglion → rostral solitary nucleus → VPM thalamus → gustatory cortex is the correct pathway for anterior 2/3 tongue taste.

Lecture 14 — Sensory Receptors & Neurotransmitters (35 questions)

Q120. A woman presented with depressive symptoms and suicidal thoughts. Which of the following neurotransmitters is abnormal in this...

Correct answer: C. Serotonin

Serotonergic dysfunction is classically implicated in depression, hence first-line SSRI treatment.

Q121. What is the mechanism of action of botulinum toxin? [NL 2017]

Correct answer: C. Blocks presynaptic fusion of neurotransmitter vesicle

Botulinum toxin cleaves SNARE proteins, blocking presynaptic ACh vesicle fusion.

Q122. An 80-year-old man had a history of progressive short-term memory loss and problems with words in speaking and writing. He was...

Correct answer: E. Blocking transporters for dopamine reuptake

This is the FALSE statement. Dopamine reuptake blockade has nothing to do with Alzheimer's pathogenesis (that's more a Parkinson's/stimulant-pharmacology concept) — the true Alzheimer's mechanisms are amyloid plaques, neurofibrillary tangles, cortical atrophy, and increased beta-secretase activity.

Q123. A patient came to the hospital with continuous generalized tonic-clonic seizures for 15 minutes. What is the most appropriate...

Correct answer: A. Diazepam

First-line treatment for active status epilepticus is a benzodiazepine — Diazepam (A). The archive answer 'Fluoxetine' (D) is a clear error; fluoxetine is an SSRI antidepressant with no role in acute seizure termination.

♦ Corrected from the original archive extraction, which had marked D. Fluoxetine — that was wrong.

Q124. The summation of EPSPs from numerous presynaptic nerve fibers converging onto one postsynaptic neuron is called. . .

Correct answer: A. Spatial summation

Spatial summation — different presynaptic fibers, same postsynaptic neuron, inputs arriving together.

Q125. A 20-year-old college student presents with depressive symptoms, including sleep disturbance and anorexia for 1 month. This...

Correct answer: C. Inhibition of serotonin transporter

This vignette describes standard SSRI treatment for depression — mechanism is inhibition of the serotonin transporter (C). The archive answer 'Stimulation of GABA receptors' (A) is wrong (that describes a benzodiazepine-type drug, not an antidepressant for this presentation) — confirmed by the near-identical Q134, which correctly answered C for the same question.

♦ Corrected from the original archive extraction, which had marked A. Stimulation of GABA receptors — that was wrong.

Q126. A woman presents with progressive muscle weakness. PE: dysphagia, ptosis when asked to perform repeated blinking. Which of the...

Correct answer: C. Anti-acetylcholine receptor antibody

Fatigable weakness + dysphagia + ptosis worsening with repeated use = myasthenia gravis, anti-AChR antibody.

Q127. A 70-year-old patient presents with dementia and cognitive problems. He received donepezil as a treatment. Which of the...

Correct answer: D. Acetylcholine

Donepezil is an acetylcholinesterase inhibitor — it raises synaptic acetylcholine.

Q128. An 80-year-old man had history of progressive short-term memory loss and problems with words in speaking and writing. He was...

Correct answer: E. Blocking transporters for dopamine reuptake

Same false statement as Q122.

Q129. A 30-year-old woman presents with bilateral ptosis and blurred vision. A complete history reveals that the woman has also...

Correct answer: C. Inhibition of acetylcholine receptor.

Ptosis + blurred vision + generalized fatigue/weakness = myasthenia gravis — caused by inhibition/blockade of the acetylcholine receptor.

Q130. A woman presented with depressive symptoms and suicidal thoughts. Which of the following neurotransmitter is abnormal in...

Correct answer: A. Serotonin

Same as Q120.

Q131. A 30-year-old woman presents with bilateral ptosis and blurred vision. A complete history reveals that the woman has also...

Correct answer: C. Inhibition of acetylcholine receptor.

Same as Q129.

Q132. What is the mechanism of action of botulinum toxin?

Correct answer: A. Blocks presynaptic fusion of neurotransmitter vesicle

Same as Q121.

Q133. A 45-year-old male presents to the emergency department with a deep wound on his foot following a gardening accident. He...

Correct answer: D. Block inhibitory interneurons

This is tetanus (Clostridium tetani, lockjaw, muscle spasms after a contaminated wound) — tetanospasmin travels retrogradely to the CNS and blocks release of inhibitory neurotransmitters from spinal inhibitory interneurons, causing disinhibited, excessive motor firing and rigidity. (Note option A's text is garbled in the source — likely intended as a distractor describing something else entirely, not the correct mechanism.)

Q134. A 20-year-old college student presents with depressive symptoms, including sleep disturbance and anorexia for 1 month. This...

Correct answer: C. Inhibition of serotonin transporter

Same SSRI mechanism as Q125, correctly extracted here — inhibition of the serotonin transporter.

Q135. A woman presents with auditory hallucination and delusion. Which of the following neurotransmitters is most likely...

Correct answer: C. Dopamine

Auditory hallucinations + delusions = psychosis, classically linked to excess dopaminergic activity.

Q136. A 70-year-old patient presents with dementia and cognitive problems. He received donepezil as a treatment. Which of the...

Correct answer: C. Acetylcholine

Same as Q127 — donepezil affects acetylcholine.

Q137. An 80-year-old man had history of progressive short-term memory loss and problems with words in speaking and writing. He was...

Correct answer: E. Blocking transporters for dopamine reuptake

Same false statement as Q122/128.

Q138. A patient receiving chemotherapy experiencing nausea and vomiting which antiemetic medication should be administered that...

Correct answer: A. 5-HT₃ receptor

Chemotherapy-induced nausea/vomiting is strongly mediated by 5-HT₃ receptors; ondansetron (a 5-HT₃ antagonist) is first-line.

Q139. Which stimuli (agents) do not stimulate the nociceptor?

Correct answer: E. Vibration

Vibration is a mechanical stimulus (detected by mechanoreceptors), not a chemical inflammatory mediator — the other four are all part of the classic 'inflammatory soup' that sensitizes/activates nociceptors.

Q140. What is the term for the time between the arrival of an action potential at a synaptic terminal and its effect on the...

Correct answer: A. synaptic delay

Synaptic delay is the ~0.5-1ms gap between presynaptic AP arrival and the postsynaptic effect, reflecting the time cost of vesicle fusion/diffusion/receptor binding.

Q141. Which receptor is ionotropic?

Correct answer: A. AMPA

AMPA is an ionotropic (ligand-gated ion channel) glutamate receptor. mGluR6, GABA-B, and 5-HT1 are all metabotropic.

Q142. What is the mechanism of fluoxetine?

Correct answer: A. selective serotonin uptake inhibitors

Fluoxetine is an SSRI, blocking SERT.

Q143. What is the mechanism of the drug ondansetron?

Correct answer: C. Block peripheral and central 5-HT3 receptor

Ondansetron blocks 5-HT3 receptors.

Q144. What is the mechanism of haloperidol?

Correct answer: B. Block D2 receptor

Haloperidol is a typical antipsychotic, blocking D2 dopamine receptors.

Q145. A patient got splashed by heated oil during cooking. Which of the following is the receptor (channel) for heat in this case?

Correct answer: A. TRPV2

Heated oil (extreme noxious heat) is detected by TRPV2 (A), the higher-threshold heat-nociception channel. The archive answer 'TRPM8' (C) is clearly wrong — TRPM8 is the COLD/menthol receptor, the opposite end of the thermal spectrum from what this scenario describes.

♦ Corrected from the original archive extraction, which had marked C. TRPM8 — that was wrong.

Q146. A woman with fatigue, social withdrawal, and depression likely has an issue with which neurotransmitter?

Correct answer: C. Serotonin

Same depression-serotonin link as Q120/130.

Q147. What does Fluoxetine inhibit?

Correct answer: A. SERT

Fluoxetine inhibits SERT (serotonin transporter).

Q148. A 75-year-old woman with progressive memory loss and cognitive decline received memantine and galantamine as treatment. Which...

Correct answer: B. Acetylcholine

Galantamine is a cholinesterase inhibitor (same drug class purpose as donepezil) — it affects acetylcholine.

Q149. Which of these is not an ionotropic receptor?

Correct answer: C. GABA B Receptor

GABA-B is metabotropic (G-protein-coupled) — the odd one out among these ionotropic receptors (AMPA, nAChR, 5-HT3 are all ionotropic).

Q150. All of the following are G-protein-coupled receptors EXCEPT?

Correct answer: E. 5-HT3 of Serotonin

5-HT3 is the one ionotropic serotonin receptor subtype — every other listed receptor (D2, muscarinic ACh, mGluR6, 5-HT1) is a genuine GPCR.

Q151. A patient is treated with Donepezil and Memantine for Alzheimer's disease. What is the effect of Memantine?

Correct answer: C. Glutamate (NMDA receptor antagonist)

Memantine is an NMDA receptor antagonist.

Q152. Which receptor is primarily blocked by haloperidol?

Correct answer: A. Dopamine D2 receptor

Haloperidol primarily blocks D2 dopamine receptors.

Q153. Loss of smell and taste (anosmia and ageusia) can result from a deficiency in which receptor, also known for its role in viral...

Correct answer: C. ACE2

ACE2 is expressed on olfactory support cells and is the same receptor SARS-CoV-2 uses for cell entry — plausibly linking to anosmia/ageusia.

Q154. Which of the following neurotransmitter-transporter pairs is correct?

Correct answer: A. VGAT - GABA

VGAT loads GABA into synaptic vesicles — a correctly matched pair. The other four options mismatch a transporter/enzyme with the wrong neurotransmitter.

Lecture 15 — Visual System (27 questions)

Q155. A patient exhibits the near vision reflex defect. Which of the following DOES NOT the characteristic of near vision reflex?

Correct answer: B. Levator palpebrae superioris contraction

This is the one NOT part of the near-reflex triad. The near triad is convergence + pupillary constriction + accommodation (lens convexity) — lid elevation (levator palpebrae) plays no role.

Q156. A patient exhibits the near vision reflex defect. Which of the following nerves involves the near vision reflex?

Correct answer: A. CN. III and parasympathetic nervous system

The near reflex is mediated by CN III (parasympathetic fibers for accommodation/constriction, plus the somatic fibers driving convergence via the medial recti).

Q157. A patient has a tumor compression at the optic chiasma. Which of the signs and symptoms would be found in this patient?

Correct answer: C. Bitemporal heteronymous hemianopia

Optic chiasm compression classically damages crossing nasal-retina fibers bilaterally, producing bitemporal hemianopia (loss of both temporal visual fields) — the classic pituitary-tumor presentation.

Q158. Which of the following nerve fibers involves this near vision reflex?

Correct answer: C. Parasympathetic nerve fiber

Parasympathetic fibers (via CN III) drive the pupillary constriction and accommodation components of the near reflex.

Q159. Which of the following structures would likely damage after examination the light reflex and the result as follow the table below?

Archive answer: A. Rt. CN. III

FLAGGED — this question references a results table that isn't included in the extracted text, so I can't independently verify the specific finding pattern against the given options. Left as originally marked.

■ Flagged as genuinely ambiguous, image/table-dependent, or having more than one defensible answer — verify against your source material.

Q160. Which of the following statements is true regarding visual systems?

Correct answer: D. Information from the right sides of visual fields finally synapses with the left primary visual area.

Information from the right visual field of both eyes ultimately synapses in the LEFT primary visual cortex (visual pathways cross so that each hemisphere processes the opposite/contralateral visual field).

Q161. Which part of the retina has no photoreceptive cells?

Correct answer: D. Optic disc

The optic disc has zero photoreceptors — it's the anatomical basis of the physiologic blind spot.

Q162. When light shines into the right eye, the right pupil has no response, but the left pupil constricts. When light shines into...

Correct answer: C. Right oculomotor nerve

The right pupil fails to respond in BOTH scenarios (whether the right or left eye is stimulated) — meaning the right pupil's EFFERENT pathway (right CN III/oculomotor) must be damaged, since the left eye's intact afferent pathway should still trigger a consensual right-pupil response if the right efferent were working. The left pupil constricting normally in both cases confirms the left afferent and efferent pathways are both intact.

Q163. What structure in the eye absorbs light?

Correct answer: A. Choroid

The pigmented choroid absorbs light, preventing internal scatter/reflection.

Q164. What condition is indicated by an Intraocular Pressure (IOP) > 25mmHg?

Correct answer: D. Glaucoma

IOP > 21 mmHg (and especially >25) is the hallmark finding of glaucoma.

Q165. Which of the following correctly arranges the layers of the retina from the Rod and Cone cell layer to the optic nerve fiber...

Correct answer: C. Rods and cones layer → Outer nuclear layer → Inner nuclear layer → Ganglion cell layer → Optic nerve fiber layer

Correct order from photoreceptor toward the optic nerve: rods/cones → outer nuclear layer (photoreceptor cell bodies) → inner nuclear layer (bipolar/horizontal/amacrine cells) → ganglion cell layer → optic nerve fiber layer.

Q166. What is the result of damage to the primary somesthetic sensory area?

Correct answer: E. Anesthesia

Damage to the primary SOMATOSENSORY (somesthetic) area causes loss of sensation — anesthesia — in the corresponding body region, not motor or visual deficits.

Q167. A patient presents with a left pupil that does not constrict and is larger than the right pupil, with a history of lower...

Archive answer: D. CN III

FLAGGED — the pupil finding (dilated, non-reactive, larger pupil) clearly points to CN III (D), but the vignette also mentions 'lower facial weakness,' which CN III doesn't explain (that's a CN VII finding, a different level/nerve entirely). None of the given options (optic tract/nerve/chiasm/CN III/optic radiation) cleanly account for BOTH findings together — CN III remains the best fit for the primary (pupil) finding, but the facial weakness detail doesn't fit any listed option well. Possible imprecision in how the vignette was written.

■ Flagged as genuinely ambiguous, image/table-dependent, or having more than one defensible answer — verify against your source material.

Q168. Surface ectoderm gives rise to....?

Correct answer: A. lens

The lens is the classic surface ectoderm derivative (via the lens placode).

Q169. Which of the following specializations of the fovea increases visual acuity?

Correct answer: C. lateral displacement of ganglion cell

At the fovea, inner retinal layers are displaced laterally, letting light reach the densely packed cones with minimal obstruction — the structural basis of maximal visual acuity there.

Q170. Damage to which nerve affects the near vision reflex?

Correct answer: A. CN.III

CN III damage affects the near reflex (accommodation, pupillary constriction, and convergence via the medial recti).

Q171. What structure is damaged in Right Homonymous Hemianopia?

Correct answer: A. Left optic tract

A RIGHT homonymous hemianopia results from a lesion of the LEFT post-chiasmal visual pathway — most simply, the left optic tract.

Q172. Which of the following abnormal cells cause the most color blindness? [

Correct answer: C. Abnormal cone cells

Color vision depends on the three cone photopigment types — most color blindness results from cone opsin gene defects.

Q173. What is the effect of sympathetic fibers on vision? [

Correct answer: C. It causes the pupil to dilate.

Sympathetic activation (dilator pupillae, alpha-1) causes pupillary dilation.

Q174. What is not correct about the near vision reflex?

Correct answer: B. Levator palpebrae superioris contraction

Same near-reflex-triad fact as Q155 — lid elevation isn't part of it.

Q175. What is the effect of retinal detachment?

Correct answer: A. No blood supply to photoreceptor cells

Retinal detachment cuts off the outer retina (photoreceptors) from the choroidal blood supply they depend on via diffusion.

Q176. Reflex test using wisp of cotton to touch sclera, with results as shown in the table. Where is the problem? (Table: Stimulate...

Correct answer: A. Left Trigeminal

Touching the RIGHT sclera produces normal bilateral blinking (afferent and both efferents intact on that side), but touching the LEFT sclera produces NO blink in either eye — since even the direct (ipsilateral) response fails, the problem must be in the LEFT afferent pathway, i.e. the left trigeminal nerve (CN V, sensory limb of this reflex).

Q177. What is the visual field defect in a lesion of the right Meyer loop?

Correct answer: A. Left homonymous superior quadrantanopia

Meyer's loop carries fibers for the CONTRALATERAL SUPERIOR visual field; a right-sided lesion causes a left, superior quadrant defect ('pie in the sky').

Q178. Arrange the following events in order:

Archive answer: B. Closure of Na channel

This is an ORDERING question, not a single-best-answer question, so a single letter doesn't capture the answer. The correct phototransduction sequence is: D (structural change of the photopigment / rhodopsin bleaching) -> A (activation of transducin) -> C (activation of phosphodiesterase) -> E (decreased intracellular cGMP) -> B (closure of the Na⁺ channel, hyperpolarizing the photoreceptor). The archive's single letter 'B' likely reflects only the LAST step of this cascade, not the full requested order.

■ Flagged as genuinely ambiguous, image/table-dependent, or having more than one defensible answer — verify against your source material.

Q179. A lesion in the right Meyer's loop (part of the optic radiation) causes what visual field defect?

Correct answer: A. Left homonymous superior quadrantanopia

Same Meyer's loop fact as Q177.

Q180. A patient with a right temporal lobe occlusion is likely to have which visual field defect?

Correct answer: C. Superior left quadrantanopia

Right temporal lobe (Meyer's loop) damage causes left, superior quadrantanopia.

Q181. During a corneal reflex test, touching the right cornea causes no blink in either eye, while touching the left cornea causes...

Correct answer: C. Right trigeminal nerve

Touching the RIGHT cornea produces no blink in either eye (afferent failure), while touching the LEFT cornea produces normal bilateral blinking — localizing to the right trigeminal nerve (afferent limb).

Lecture 16 — Auditory & Vestibular Systems (21 questions)

Q182. Which of the following items is found at the level of rostral pons?

Correct answer: D. Motor and Principle nuclei of CN. V

The motor and principal sensory nuclei of CN V are a classic rostral pons landmark. The archive answer 'Inferior vestibular nucleus' (E) is inconsistent with Q184 (essentially the same question, 'rostral-mid pons'), which correctly answered D with the identical option list — that internal inconsistency points to E being an extraction error here. The inferior vestibular nucleus is more properly associated with the pontomedullary junction/medulla, not rostral pons.

◆ Corrected from the original archive extraction, which had marked E. Inferior vestibular nucleus — that was wrong.

Q183. A patient has facial nerve damage resulting in hyperacusis (). Which of the following would likely be interrupted in the...

Correct answer: D. Stapedius reflex is interrupted.

The stapedius muscle (which dampens loud sound) is innervated by CN VII — facial nerve damage disables this reflex, causing hyperacusis.

Q184. Which of the following items is found at the level of rostral-mid pons?

Correct answer: D. Motor and Principle nuclei of CN. V

CN V motor/principal sensory nuclei sit at the rostral-mid pons level.

Q185. Which of the following vestibular nuclei gives the axons to the oculomotor nucleus? (only 4 choices)

Correct answer: C. Superior vestibular

The superior vestibular nucleus is the main source of ascending vestibulo-ocular connections (via the MLF) to the oculomotor nucleus.

Q186. Which structures release sound waves from the inner ear?

Correct answer: C. Round window

The round window bulges outward to release the pressure wave transmitted through the cochlear fluids.

Q187. Which of the following statements is true regarding sound stimulation?

Correct answer: C. Inner hair cells vibrate to stimulate the tectorial membrane and cause an action potential, which carries through the peripheral process of the spinal ganglion.

Inner hair cell stereocilia deflection against the tectorial membrane generates the receptor potential carried by the peripheral processes of spiral ganglion neurons (the source text says 'spinal ganglion,' almost certainly a typo for 'spiral ganglion,' the correct cochlear structure).

Q188. What structure is used to detect movement when using an elevator (vertical acceleration)?

Correct answer: D. Saccule

Vertical linear acceleration (an elevator starting/stopping) is detected by the saccule (D), whose macula is oriented in a vertical plane. The archive answer 'Superior semicircular canal' (A) is wrong — semicircular canals detect ANGULAR/rotational acceleration (like nodding), not linear acceleration.

♦ Corrected from the original archive extraction, which had marked A. Sup semicircular — that was wrong.

Q189. What structure is primarily used to detect the motion of tilting the head up (vertical linear movement)?

Correct answer: B. Saccule

Same vertical-linear-acceleration logic — the saccule.

Q190. What structure is primarily used to detect the motion of nodding the head forward?

Correct answer: D. Superior semicircular canal

Nodding (a rotational, sagittal-plane movement) is sensed by the superior (anterior) semicircular canal.

Q191. When you extend your head back (posterior), what receptor is stimulated?

Correct answer: A. Sup semicircular

Extending the head backward is the same rotational plane as nodding forward — same canal (superior/anterior) detects both directions.

Q192. What is the location of the vomiting center?

Correct answer: C. Area postrema

Area postrema (a circumventricular organ with a leaky BBB) functions as the chemoreceptor trigger zone for vomiting.

Q193. What structure is used to reduce surrounding sound (by inhibiting outer hair cells) to better listen to people talk?

Correct answer: D. Superior olivary complex

Selectively inhibiting outer hair cells to improve signal-to-noise for speech is the job of the medial olivocochlear efferent system, which originates in the superior olivary complex (D). The archive answer 'Stapedius muscle' (A) instead describes a different, non-selective protective mechanism (dampening ossicle movement for loud sounds overall), not the outer-hair-cell-specific 'listen to speech in noise' mechanism this question describes.

♦ Corrected from the original archive extraction, which had marked A. Stapedius muscle — that was wrong.

Q194. What is stimulated when walking forward?

Correct answer: D. macula utricle

Walking forward involves HORIZONTAL linear acceleration/deceleration, detected by the utricle (D), not the saccule (E, the archive's answer, which detects VERTICAL linear acceleration like an elevator).

♦ Corrected from the original archive extraction, which had marked E. macula saccule — that was wrong.

Q195. Trouble hearing low frequency suggests a possible occlusion where?

Correct answer: B. Apex of Basilar membrane

Low frequencies are encoded at the cochlear apex (wide, flexible basilar membrane).

Q196. Which pathway of hearing is correct?

Correct answer: A. Cochlear nerve > Cochlear nu. > Sup. Olivary > Medial Geniculate body

Cochlear nerve -> cochlear nucleus -> superior olivary complex -> (lateral lemniscus ->) medial geniculate body is the correct early auditory pathway sequence.

Q197. An 80-year-old patient presents with presbycusis hearing loss (50-8,000HZ). Which of the following is the most likely damaged?

Correct answer: B. Outer hair cell

Presbycusis classically results from cumulative outer hair cell damage, especially at the high-frequency-encoding cochlear base.

Q198. A man slipped and was about to fall. He extended his upper and lower extremity. Which structure involved this reflex?

Correct answer: B. Lateral vestibulospinal tract

The lateral vestibulospinal tract facilitates extensor/antigravity limb tone — exactly the reflex needed to extend limbs and catch a fall.

Q199. What causes Presbycusis? [

Correct answer: C. Damage to hair cells

Presbycusis is fundamentally sensorineural — cumulative hair cell damage, not a conductive (ossicle/tympanic membrane) problem.

Q200. What is incorrect about stimulating the crista ampullaris? [

Correct answer: D. It detects linear acceleration of the head.

This is the FALSE statement (matching 'incorrect'). The crista ampullaris detects ANGULAR acceleration, not linear — linear acceleration is the otolith organs' (utricle/sacculle) job.

Q201. Which tract can inhibit the vestibular reflex to the limb muscles?

Correct answer: B. Corticospinal tract

The corticospinal tract provides descending voluntary control that can override/modulate the more automatic vestibulospinal postural reflexes.

Q202. A test for presbycusis is done. A patient aged 45/60/90 starts hearing at which level?

Correct answer: B. 25-50 dB

Presbycusis progresses gradually, with threshold shifts typically beginning in the mild-to-moderate 25-50 dB range.

Mixed / Other Topics (49 questions)

Q203. A lumbar puncture consists of a needle being inserted through a space into the lumbar cistern, and aims to retrieve a sample...

Correct answer: B. L4-L5

The spinal cord ends around L1-L2 in adults, so lumbar puncture is performed lower (L3-L4 or L4-L5) to avoid the cord.

Q204. Space that connects between the lateral and the third ventricles is called . . .

Correct answer: A. interventricular foramen

The interventricular foramen (of Monro) connects each lateral ventricle to the third ventricle.

Q205. Which of the following types of nerve fibers has the fastest conduction velocity?

Correct answer: E. A- α (Ia) fibers

A-alpha (Ia) fibers are the largest-diameter, most heavily myelinated, and fastest-conducting peripheral fibers.

Q206. The labyrinthine artery is an important source of blood supply to the inner ear. Which of the following arteries represents...

Correct answer: A. Anterior inferior cerebellar

The labyrinthine artery most commonly arises from the anterior inferior cerebellar artery (AICA) — which is also why AICA territory strokes classically affect hearing/balance alongside facial nerve function (lateral pontine syndrome).

Q207. Which of the following would be expected to see signs and symptoms of the patient from the lesion below?

Archive answer: E. Abnormally conjugate eye movement to left with nystagmus to the right sides.

FLAGGED — this question references a lesion diagram ('the lesion below') not included in the extracted text, so I cannot independently verify the finding against the actual image. Left as originally marked.

■ Flagged as genuinely ambiguous, image/table-dependent, or having more than one defensible answer — verify against your source material.

Q208. Which of the following sensation losses on the ipsilateral side of the lesion and at the only the level L2?

Archive answer: E. Fine touch

FLAGGED, moderate confidence the archive is wrong. At the EXACT segmental level of a Brown-Séquard hemisection, there's a classically-taught narrow ipsilateral band of pain/temperature loss specifically — because those entering dorsal root pain fibers ascend 1-2 segments ipsilaterally in Lissauer's tract before crossing, so right at the level of the cut they haven't crossed yet. This makes 'Pain' (A) the more specifically-taught answer for this exact 'at the level only' phenomenon, rather than 'Fine touch' (E, the archive's answer). Worth checking your slide's specific framing, since this is a genuinely subtle point.

■ Flagged as genuinely ambiguous, image/table-dependent, or having more than one defensible answer — verify against your source material.

Q209. A patient was diagnosed with meningitis by lumbar puncture. Which of the following sites is the most commonly used for CSF...

Correct answer: D. L4-L5

Same LP-level logic as Q203.

Q210. Which of the following receptor type receptors detect the tension of muscle and tendon? (only 4 choices)

Correct answer: C. Proprioceptor

Proprioceptors (e.g., muscle spindles, Golgi tendon organs) detect muscle/tendon tension and length.

Q211. Which of the following region is the most common of the encephalocele?

Correct answer: C. Occipital

Encephalocele is a CRANIAL defect (brain/meninges herniating through a skull defect) — 'Lumbar' (A, the archive's answer) isn't even an anatomically valid location for a cranial encephalocele (a spinal defect at that level would instead be called a meningocele). The most common encephalocele location, especially in Western populations, is occipital (C).

♦ Corrected from the original archive extraction, which had marked A. Lumbar — that was wrong.

Q212. A 43-year-old woman was sent to consult the doctor with drooping of right side of mouth, could not smile symmetrically....

Correct answer: A. Internal capsule

Forehead-sparing facial weakness (can still close eyes tightly and raise eyebrows, but the lower face droops) is the classic UMN facial palsy pattern, since the upper face receives bilateral cortical innervation. Internal capsule is a classic site for this kind of corticobulbar fiber lesion (e.g., in stroke).

Q213. A 65-year-old man has a pernicious anemia and memory problem. Which of the following compounds in blood is higher than normal...

Correct answer: D. homocysteine

B12 deficiency (as in pernicious anemia) elevates homocysteine (and methylmalonic acid) because B12 is a required cofactor for converting homocysteine back to methionine — so methionine would actually be LOW, not elevated, ruling that option out.

Q214. If the right Edinger Westphal nucleus is damaged from whatever, what is (are) the testing results?

Correct answer: B. When light shines into Rt. eye, Lt. pupil constricts, but Rt. not respond.

With the right Edinger-Westphal nucleus (right CN III's parasympathetic output) damaged, the right pupil can never constrict — whether light is shone in the right eye (direct response fails) or the left eye (consensual response to the right pupil also fails, since the efferent side is broken) — while the left pupil, with intact efferent supply, constricts normally either way. This matches: light in right eye -> left pupil constricts, right does not.

Q215. Which fibers carry the first pain after injecting a needle into the skin? (only 4 choices)

Correct answer: B. Aδ (delta)

First/fast pain (from a needle prick) is carried by A-delta fibers (B). The archive answer 'C fibers' (A) is wrong — C fibers carry SECOND/slow, dull pain, not the initial sharp sensation.

♦ Corrected from the original archive extraction, which had marked A. C fibers — that was wrong.

Q216. Which of the following types of nerve fibers has the fastest conduction velocity?

Correct answer: E. A-α (Ia)

Same as Q205 — A-alpha (Ia) fibers are fastest.

Q217. What type of transport is involved in the rabies virus, and by which specific molecular motor is it mediated?

Correct answer: C. Retrograde - dynein mediated

Rabies uses retrograde axonal transport (toward the cell body), driven by the motor protein dynein.

Q218. Which of the following conditions is primarily characterised by the presence of plaques in the periventricular regions?

Correct answer: D. Multiple sclerosis

Periventricular white-matter plaques are the hallmark of multiple sclerosis (D), not Alzheimer's disease (A, the archive's answer — Alzheimer's pathology is amyloid plaques diffusely in gray matter, a completely different finding). This is confirmed by Q251, an essentially identical question later in this same archive, which correctly answered D (Multiple Sclerosis) for the same finding.

♦ Corrected from the original archive extraction, which had marked A. Alzheimer's disease — that was wrong.

Q219. What is the type of synapse between an axon and another axon called?

Correct answer: C. Axo-axonic

Axon-to-axon synapse = axo-axonic, typically for presynaptic modulation.

Q220. What is the percent of the body's glucose used by the brain?

Correct answer: A. 25%

The brain uses roughly 20-25% of the body's glucose despite being ~2% of body weight.

Q221. What causes glutamate toxicity to happen?

Correct answer: A. hypoglycemia

Hypoglycemia (energy failure) impairs the ATP-dependent glutamate reuptake/glutamine synthetase system, letting extracellular glutamate build up and drive excitotoxicity.

Q222. What would be the result of a lesion to the Corticospinal tract in the pons?

Archive answer: A. motor loss on the right side

FLAGGED — the extracted question stem doesn't specify which side the pontine lesion is on, which is essential information for this question type (since a lesion above the pyramidal decussation, like one in the pons, causes CONTRALATERAL motor loss). Without knowing the lesion's side from the original stem/image, I can't independently confirm whether 'right side' is the correct paired answer.

■ Flagged as genuinely ambiguous, image/table-dependent, or having more than one defensible answer — verify against your source material.

Q223. Which tract does not pass through the inferior cerebellar peduncle?

Correct answer: C. pontocerebellar

Pontocerebellar fibers travel through the MIDDLE cerebellar peduncle, not the inferior one — correctly excluded here.

Q224. Mechanoreceptors of the skin transmit signals via which type of nerve fiber?

Correct answer: A. A beta

Cutaneous mechanoreceptors signal via A-beta fibers.

Q225. What doesn't synapse at the spinal trigeminal nucleus?

Correct answer: E. pain temp from accessory

CN XI (accessory) is purely motor and contributes no sensory fibers to the spinal trigeminal nucleus — consistent with the Q90/Q95 fix above.

Q226. Which of the following is an epicritic sensation?

Correct answer: C. fine touch

Fine touch is epicritic (discriminative); pain and temperature are protopathic/affective.

Q227. What receptor is responsible for the sensory adaptation experienced when wearing a necklace or watch?

Correct answer: B. Pacinian receptor

Same rapid-adaptation fact as Q3.

Q228. What transports glutamate into a glial cell?

Correct answer: A. EAAT

EAAT transports glutamate into glial cells (astrocytes).

Q229. Transportation of Rabies occurred as which of the following directions and by which specific molecular motor?

Correct answer: C. Retrograde - dynein mediated

Same as Q217 — retrograde, dynein-mediated.

Q230. Which part of the brain specializes for motor control of our body, limb and eye movement?

Correct answer: C. Frontal cortex

The frontal cortex (primary motor cortex, frontal eye fields) specializes in motor control of body, limb, and eye movement.

Q231. Which of the following non-CNS structures is located in the Longitudinal fissure?

Correct answer: B. falx cerebri

The falx cerebri (a dural fold, non-neural connective tissue) sits within the longitudinal fissure between the cerebral hemispheres.

Q232. In normal condition, which of the following is used by a neuron to produce ATP?

Correct answer: A. glucose

Under normal conditions, glucose is the primary neuronal ATP substrate (lactate becomes more prominent specifically during high demand or limited glucose delivery).

Q233. What is the condition where meninges and brain tissue protrude?

Archive answer: A. Cranium bifidum with meningoencephalocele

FLAGGED, moderate confidence. 'Meninges and brain tissue' protruding (without mention of a herniated ventricle/CSF space) most precisely matches meningoencephalocele (C) — meninges + encephalon (brain). The archive answer 'meningoencephalocele' (A) additionally implies herniation of a ventricle, a detail not mentioned in the question stem. Worth checking your slide's exact terminology.

■ Flagged as genuinely ambiguous, image/table-dependent, or having more than one defensible answer — verify against your source material.

Q234. Which reflex test is appropriate for the C5-6 segment?

Archive answer: E. Brachioradialis jerk

FLAGGED — both biceps (C) and brachioradialis (E) reflexes are classically taught as C5-C6 level reflexes; depending on how your specific course splits the segmental levels (some sources emphasize biceps=C5, brachioradialis=C6), either could be defensible as 'the' answer for 'C5-6.' Left as originally marked.

■ Flagged as genuinely ambiguous, image/table-dependent, or having more than one defensible answer — verify against your source material.

Q235. Neuroblasts from the alar plate can migrate ventrally to form which of the following structures?

Correct answer: B. Olivary nucleus

The inferior olivary nucleus is a classic example of alar-plate-derived neuroblasts migrating ventrally.

Q236. What disease is characterized by the absence of iris? .

Correct answer: C. Aniridia

Absence of the iris is literally the definition of aniridia (C) — 'an-' (without) + '-iridia' (iris). The archive answer 'Coloboma' (A) is wrong; a coloboma is a partial notch/gap defect, not complete absence.

♦ Corrected from the original archive extraction, which had marked A. Coloboma — that was wrong.

Q237. A 58-year-old patient has a drooping left eyelid and cannot wrinkle their forehead. What is the cause of the lesion?

Correct answer: B. motor cn7

Drooping eyelid AND inability to wrinkle the forehead together (forehead involved, unlike Q212) is the LMN/peripheral facial palsy pattern (Bell's palsy) — CN VII itself, not a central lesion.

Q238. What structure is primarily stimulated when running forward (horizontal linear acceleration)?

Correct answer: A. Utricle

The original question stem and options were illegible/redacted in the source archive, so this question has been fully reconstructed. Running forward involves horizontal linear acceleration/deceleration with each stride, detected by the utricle (whose macula lies in the horizontal plane) — the same structure tested by 'driving a car' style questions elsewhere in this set.

♦ Corrected from the original archive extraction, which had marked None. (unmarked/illegible) — that was wrong.

Q239. All statements are true about corneal epithelium 'except'

Correct answer: D. has a rich blood supply

This is the FALSE statement. The cornea is avascular — nourished by diffusion from tears and aqueous humor, not blood vessels — which is exactly why 'has a rich blood supply' is the incorrect statement being asked for.

Q240. Which nerve fiber is most sensitive to anesthesia?

Correct answer: D. B fiber

B fibers (small-diameter, lightly myelinated autonomic preganglionic fibers) are classically the most sensitive to local anesthetic blockade, blocked even before unmyelinated C fibers in the typical differential-block sequence.

Q241. A 25-year-old man came to the hospital. He dropped a pan on his legs (splashed with hot water). Which receptor is involved?

Correct answer: B. TRPV1

Hot water (moderate-to-severe noxious heat) is the classic activation range for TRPV1 (B), which is also the capsaicin receptor. The archive answer 'TRPA1' (A) is wrong — TRPA1 is more associated with noxious cold and certain irritant chemicals (like mustard oil), not primarily heat.

♦ Corrected from the original archive extraction, which had marked A. TPRA1 — that was wrong.

Q242. A baby boy has a certain condition. The mother has no history of taking folic acid supplements and has no other abnormalities....

Archive answer: A. Holoprosencephaly

FLAGGED, moderate confidence the archive is wrong. Folic acid deficiency is most classically and directly linked to neural tube defects — and 'Merocephaly' (E) is a synonym for anencephaly, established elsewhere in this same archive (see Q48). The archive answer 'Holoprosencephaly' (A) has different, less folate-specific teratogenic associations (e.g., Sonic Hedgehog pathway mutations, maternal diabetes). Encephalocele (C) is also a plausible folate-linked NTD variant, so there's some legitimate competing plausibility between C and E — but A is likely the least well-supported of the three.

■ Flagged as genuinely ambiguous, image/table-dependent, or having more than one defensible answer — verify against your source material.

Q243. What receptor is related to ptosis and double vision?

Correct answer: A. N1

Ptosis + double vision (diplopia) is the classic myasthenia gravis presentation — autoantibodies against the nicotinic ACh receptor (N1/nAChR) at the neuromuscular junction.

Q244. What is the target for Opioid drugs?

Archive answer: A. Thalamus

FLAGGED, moderate confidence the archive is wrong. While opioids do act at multiple sites (spinal dorsal horn, thalamus, etc.), the single most classic, most heavily emphasized 'textbook' target for opioid analgesia — especially regarding descending pain-modulation pathways — is the periaqueductal gray (PAG, option D). The archive answer 'Thalamus' (A) isn't unreasonable, but PAG is the more iconic answer typically tested for this exact question phrasing.

■ Flagged as genuinely ambiguous, image/table-dependent, or having more than one defensible answer — verify against your source material.

Q245. A person turns their head to the right, but their eyes remain looking forward. The left eye moves to the left, while the right...

Archive answer: A. Right oculomotor

FLAGGED — working through the eye-movement logic: the right eye can abduct normally (during a leftward head turn) but fails to adduct (during a rightward head turn) — this is the classic pattern of internuclear ophthalmoplegia (INO) from a lesion of the right MLF, not literally the right oculomotor nerve/nucleus itself. None of the given options say 'MLF,' so if your course is testing INO specifically, the intended answer may be conceptually 'right MLF' even though 'Right oculomotor' (A) is the closest available option given the muscle affected. Worth checking whether your course expects MLF-level precision here.

■ Flagged as genuinely ambiguous, image/table-dependent, or having more than one defensible answer — verify against your source material.

Q246. Which molecular motor is used for rabies transport?

Correct answer: A. Dynein

Same as Q217/229 — dynein.

Q247. What is a gated-channel control?

Archive answer: E. presynaptic inhibition

FLAGGED — the question stem itself ('what is a gated-channel control?') is quite vague/ambiguous out of context, and I don't have enough information to confidently verify or refute 'presynaptic inhibition' as the intended answer. Worth checking against the specific slide this question is drawn from.

■ Flagged as genuinely ambiguous, image/table-dependent, or having more than one defensible answer — verify against your source material.

Q248. Which of these is part of the paleocerebellum?

Correct answer: A. vermis and anterior lobe

The paleocerebellum comprises the anterior lobe and (parts of) the vermis — involved in muscle tone/proprioceptive input, distinct from the archicerebellum (flocculonodular lobe, vestibular) and neocerebellum (posterior lobe, fine coordination).

Q249. Which of the following is an affective/protopathic sensation?

Correct answer: C. Pain

Pain is the classic protopathic/affective sensation — poorly localized, emotionally charged. Fine touch and two-point discrimination are epicritic instead.

Q250. A patient with Benedikt syndrome would exhibit ipsilateral CN III palsy and what other sign?

Archive answer: D. Ipsilateral cerebellar ataxia

FLAGGED, low-moderate confidence in either direction. Classic Benedikt syndrome (red nucleus + medial lemniscus involvement) most specifically causes CONTRALATERAL involuntary movements/tremor (and sometimes contralateral sensory loss) alongside the ipsilateral CN III palsy — a finding not precisely listed among these options. 'Ipsilateral cerebellar ataxia' (D, the archive's answer) doesn't cleanly match the standard teaching either, since red-nucleus/superior-cerebellar-peduncle-related movement abnormalities in Benedikt are classically described as CONTRALATERAL, not ipsilateral. None of the given options is a clean fit — recommend double-checking this one directly against your lecture slide.

■ *Flagged as genuinely ambiguous, image/table-dependent, or having more than one defensible answer — verify against your source material.*

Q251. A 40-year-old patient has a plaque in the periventricular area seen on MRI. What is the diagnosis?

Correct answer: D. Multiple Sclerosis

Periventricular plaques are the classic MRI finding in multiple sclerosis — this question is essentially identical to Q218, confirming that Q218's archived 'Alzheimer's' answer was the error, not this one.